Algebraic Topology

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Contents

1 Bases of the Fundamental Group

1.1 Homotopy equivalence.

Definition 1.1. Let X be a topological space $f, g\colon X \to Y$ then we say that f is *homotopic* to g if there is a function $H\colon X \times I \to Y$ (called a homotopy) such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x)$$

Proposition 1.2. *Homotopy is an equivalence relation.*

Lemma 1.3 (Gluing Lemma). *Suppose that $X = A \cup B$ with A, B closed and take $f_1\colon A \to Y$,*

$f_2 \colon B \to Y$. *s.t* $f_1(x) = f_2(x)$ *for all* $x \in A \cap B$. *Then*

$$f_3 \colon X \to Y, \quad f_3(x) = \begin{cases} f_1(x) & \textit{if } x \in A \\ f_2(x) & \textit{if } x \in B \end{cases}$$

is continuous.

Proof. Let C be a closed set in Y, then

$$\begin{aligned} f_3^{-1}(C) &= f_3^{-1}(C) \cap (A \cup B) \\ &= \left(f_3^{-1}(C) \cap A\right) \cup \left(f_3^{-1} \cap B\right) \\ &= f_1^{-1}(C) \cup f_2^{-1}(C) \end{aligned}$$

which is closed in X. □

Example 1.4. Any two function into $\mathbb{R}^n$ (convex set, start shaped set) are homotopic.

Indeed, take two such $f(x), g(x)$ then let

$$H \colon X \times Y \to \mathbb{R}^n, \quad H(x,t) = (1-t)f(x) + tg(x)$$

Note: we call this *the straight line homotopy*.

Relative Homotopy

Definition 1.5. Let $f, g \colon X \to Y$ with $A \subseteq X$ and $f_{|A} = g_{|A}$ then we say that f is homotopic to g relative to A if there exists a homotopy $H \colon X \times I \to Y$ between f and g

$$H(x,t) = f(x) = g(x) \quad \forall t \in I, \, \forall x \in A$$

Remark 1.6. Homotopy is a special kind of relative homotopy (for $A = \varnothing$).

Proposition 1.7. *Relative homotopy is an equivalence relation.*

Definition 1.8. Let X a topological space and let $x_1, x_2 \in X$ then a path in X going from x_1 to x_2 is a function $\gamma \colon I \to X$ s.t

$$\gamma(0) = x_1, \quad \gamma(1) = x_2$$

Definition 1.9. Suppose that γ_1 and γ_2 are two paths in X whose start point and end point coincide, then we say that γ_1 is path homotopic to γ_2 if γ_1 is homotopic to γ_2 relative to $\{0, 1\}$.

Intuitively this means that we can deform γ_1 into γ_2 without moving the endpoints. i.e $H \colon I \times I \to X$ s.t

$$H(s,t) = \begin{cases} \gamma_1(s) & \text{if } t = 0 \\ \gamma_2(s) & \text{if } t = 1 \\ \gamma_1(0) = \gamma_2(0) & \text{if } s = 0 \\ \gamma_1(1) = \gamma_2(1) & \text{if } s = 1 \end{cases}$$

Corollary 1.10. *Path homotopy is an equivalence relation.*

Definition 1.11. $\gamma_1, \gamma_2 \colon I \to X$ with

$$\gamma_1(1) = \gamma_2(0)$$

we define the *product* of these paths to be the path $\gamma_1 \cdot \gamma_2$ given by

$$\gamma_1 \cdot \gamma_2(s) = \begin{cases} \gamma(2s) & \text{if } 0 \leq s \leq 1/2 \\ \gamma(2s-1) & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

Note that this is continuous by the Gluing Lemma.

Proposition 1.12. *Path multiplication is compatible with path homotopy; i.e if*

$$\gamma_1 \sim \gamma_1', \quad \gamma_2 \sim \gamma_2'$$

then

$$\gamma_1 \cdot \gamma_2 \sim \gamma_1' \sim \gamma_2'$$

Proof. Let $H_1 \colon I \times I \to X$ and $H_2 \colon I \times I \to X$ be the corresponding homotopies then define $H \colon I \times I \to X$ by

$$H(s,t) = \begin{cases} H_1(2s,t) & \text{if } 0 \leq t \leq 1/2 \\ H_2(2s-1,t) & \text{if } 1/2 \leq t \leq 1 \end{cases}$$

We have to check that this a valid path homotopy, TODO. □

1.2 The Fundamental group

Definition 1.13. A loop in X based at x_0 is a path whose end points are both x_0.

Theorem 1.14. *Take X a top space and $x_0 \in X$ then the set of path homotopy equivalence classes of the loops based at x_0 is a group where multiplication, identity and inverses are defined as such:*

- $[\gamma_1] \cdot [\gamma_2] = [\gamma_1 \cdot \gamma_2]$
- *Identity is $[x_0]$*
- *Inverse of $[\gamma]$ is $[\overline{\gamma}]$ where*

$$\overline{\gamma}(s) = \gamma(1-s)$$

This group is called the Fundamental group of X based at x_0 denoted by $\pi_1(X, x_0)$.

Remark 1.15. Note that multiplication is well defined, since homotopy behaves nicely with path product.

Lemma 1.16. *Let $\varphi \colon I \to I$ s.t $\varphi(0) = 0$ and $\varphi(1) = 1$. Then for any path γ we have that*

$$\gamma\varphi \sim \gamma$$

we call $\gamma \circ \varphi$ a reparametrization of γ.

Proof. Let $H(s,t) = \gamma((1-t)s + t\varphi(s))$, then clearly

$$H(s,0) = \gamma(s), \quad H(s,1) = \gamma(\varphi(s))$$

$$H(0,t) = \gamma(0), \quad \gamma(1,t) = \gamma(1)$$

□

Proof of the theorem.
We have to show that this satisfies the axioms of groups:

- **Associativity.** Take $\gamma_1,\ \gamma_2,\ \gamma_3 \colon I \to X$ s.t

  $$\gamma_1(1) = \gamma_2(0), \quad \gamma_2(1) = \gamma_3(0)$$

  Then we have that

  $$(\gamma_1 \cdot \gamma_2) \cdot \gamma_3(s) = \begin{cases} \gamma_1(4s) & \text{if } 0 \leq s \leq 1/4 \\ \gamma_2(4s-1) & \text{if } 1/4 \leq s \leq 1/2 \\ \gamma_3(2s-1) & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

  and

  $$\gamma_1 \cdot (\gamma_2 \cdot \gamma_3)(s) = \begin{cases} \gamma_1(2s) & \text{if } 0 \leq s \leq 1/2 \\ \gamma_2(4s-2) & \text{if } 1/2 \leq s \leq 3/4 \\ \gamma_3(4s-3) & \text{if } 3/4 \leq s \leq 1 \end{cases}$$

  We want to find $\varphi \colon I \to I$ s.t

  $$(\gamma_1 \cdot \gamma_2) \cdot \gamma_3(s) = \gamma_1 \cdot (\gamma_2 \cdot \gamma_3)(\varphi(s))$$

  Indeed, define

  $$\varphi(s) = \begin{cases} 2s & \text{if } 0 \leq s \leq 1/4 \\ s + 1/4 & \text{if } 1/4 \leq s \leq 1/2 \\ 1/2s + 1/2 & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

  Check that this is a valid reparametrization; i.e that the equality above holds.
- **Identity.** We will only show left-identity: right is completely analogous. Take $\gamma \colon I \to X$, let $x_0 = \gamma(0)$ and $x_0 \colon I \to X$ s.t $x_0(s) = x_0$. Then

  $$x_0 \cdot \gamma(s) = \begin{cases} x_0 = \gamma(0) & \text{if } 0 \leq s \leq 1/2 \\ \gamma(2s-1) & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

  Then clearly $\gamma(\varphi(s)) = x_0 \cdot \gamma(s)$ via

  $$\varphi(s) = \begin{cases} 0 & \text{if } 0 \leq s \leq 1/2 \\ 2s-1 & \text{if } 1/2 \leq s \leq 1 \end{cases}$$

Check that the reparametrization is valid.

- **Inverse.** Suppose that $\gamma\colon I \to X$ is a path, fix $\alpha \in I$. Define the path

$$\gamma_\alpha(s) = \gamma(\alpha s)$$

We want to show that

$$\gamma \cdot \overline{\gamma} \sim \gamma(0)$$

Let $H\colon I \times I \to X$ be

$$H(s,t) = \gamma_{1-t} \cdot \overline{\gamma_{1-t}}(s)$$

Then notice that

$$H(s,0) = \gamma_1\overline{\gamma_1}(s), \quad H(s,1) = \gamma_0 \cdot \overline{\gamma_0}(s) = \gamma(0)$$

and

$$H(0,t) = \gamma_{1-t}(0) = \gamma(0), \quad H(1,t) = \gamma(0)$$

Where the last equality comes from the fact that

$$H(s,t) = \begin{cases} \gamma_{1-t}(2s) & \text{if } 0 \le s \le 1/2 \\ \overline{\gamma_{1-t}}(2s-1) = \gamma_{1-t}(2-2s) = \gamma((1-t)(2-2s)) & \text{if } 1/2 \le s \le 1 \end{cases}$$

Note that H is continuous since it is clearly continuous on $[0,1/2] \times I$ and $[1/2,1] \times I$ + Gluing lemma.

□

Theorem 1.17. *Let X be a topological space and let $x_0, x_1 \in X$; take η a path from $x_0 \to x_1$. Define $\beta_\eta\colon \pi_1(X,x_1) \to \pi_1(X,x_0)$ by*

$$\beta_\eta[\gamma] = [\eta \cdot \gamma \cdot \overline{\eta}]$$

Then β_η is an isomorphism.

Proof.

- **Well defined.** Let $[\gamma_1] = [\gamma_2]$ then

$$\begin{aligned} [\gamma_1] = [\gamma_2] &\implies \gamma_1 \sim \gamma_2 \\ &\implies \eta \cdot \gamma_1 \sim \eta \cdot \gamma_2 \\ &\implies \eta \cdot \gamma_1 \cdot \overline{\eta} \sim \eta \cdot \gamma_2 \cdot \overline{\eta} \\ &\implies [\eta \cdot \gamma_1 \cdot \overline{\eta}] = [\eta \cdot \gamma_2 \cdot \overline{\eta}] \end{aligned}$$

- **Homomorphism.**

$$\begin{aligned}\beta_\eta([\gamma_1]\cdot[\gamma_2]) &= \beta_\eta([\gamma_1\cdot\gamma_2])\\ &= [\eta\cdot\gamma_1\cdot\gamma_2\cdot\overline{\eta}]\\ &= [\eta\cdot\gamma_1\cdot x_1\cdot\gamma_2\cdot\overline{\eta}]\\ &= [\eta\cdot\gamma_1\cdot\overline{\eta}]\cdot[\eta\cdot\gamma_2\cdot\overline{\eta}]\\ &= \beta_\eta[\gamma_1]\cdot\beta_\eta[\gamma_2]\end{aligned}$$

- **Isomorphism.** Notice that

$$\begin{aligned}\beta_\eta\circ\beta_{\overline{\eta}}[\gamma] &= [\beta_\eta(\overline{\eta})\cdot\gamma\cdot\eta]\\ &= [\eta\cdot\overline{\eta}\cdot\gamma\cdot\eta\cdot\overline{\eta}]\\ &= [x_0\cdot\eta\cdot x_0]\\ &= [\gamma]\end{aligned}$$

  Similarly $\beta_{\overline{\eta}}\circ\beta_\eta$ is also identity.

□

Corollary 1.18. *If X is path connected then $\pi_1(X, x_0)$ is independent of X.*

From now on we all space will be path connected (unless stated otherwise explicitly): in this case we will simply write $\pi_1(X)$.

Definition 1.19. If $\pi_1(X) = e$ we say that X is *simply connected*

Proposition 1.20. *X is simply connected $\iff$ $\forall x_0, x_1 \in X$ any two paths between x_0 and x are homotopic.*

Proof. $\Longleftarrow$ Take $x_0 = x_1$, then any two loops are homotopic, in particular every loop is homotopic to the constant loop.

$\Longrightarrow$ Let $x_0, x_1 \in X$ and take $\gamma, \gamma' \colon x_0 \to x_1$. Note that $\gamma\cdot\overline{\gamma'}$ is a loop, therefore $\gamma\cdot\overline{\gamma'} \sim x_0$. Therefore multiplying both sides by γ' we get

$$\begin{aligned}(\gamma\cdot\overline{\gamma'})\cdot\gamma' &\sim x_0\cdot\gamma'\\ (\gamma\cdot\overline{\gamma'}) &\sim \gamma'\\ \gamma &\sim \gamma'\end{aligned}$$

□

Example 1.21. The following spaces are simply connected

- $\mathbb{R}^n$: Let γ be a loop based at 0, then let

$$H(s,t) = (1-t)\gamma(s) + t\cdot x_0 = (1-t)\gamma(s)$$

- Similarly, every convex set is simply connected.
- Similarly, every star-shaped set is simply connected.

- S^n is simply connected for $n \geq 2$ (proof later).

Notice that S^1 is not simply connected (proof later).

1.3 Homotopy of spaces

Let X, Y be topological spaces and $\varphi\colon X \to Y$ be a map. Fix $x_0 \in X$, suppose that γ is a loop based at x_0, then $\varphi(\gamma)$ is a loop based at $\varphi(x_0)$. Moreover, if $\gamma \sim \gamma'$ via homotopy H, then $\varphi(\gamma) \sim \varphi(\gamma')$ via the homotopy $\varphi \circ H$.

Proof. $H\colon I \times I \to X$ s.t

$$H(0,t) = x_0 = H(1,t), \quad H(s,0) = \gamma(s), \quad H(s,1) = \gamma'(s)$$

Then

$$\varphi \circ H\colon I \times I \to Y$$

and

$$\varphi(H(0,t)) = \varphi(x_0) = \varphi(H(1,t)), \quad H(s,0) = \varphi(\gamma(s)), \quad \varphi(H(s,1)) = \varphi(\gamma'(s))$$

□

Therefore we get a well defined map from $\pi(X, x_0)$ to $\pi(Y, \varphi(x_0))$, denote by φ_* and defined by

$$\varphi_*([\gamma]) = [\varphi \circ \gamma]$$

Proposition 1.22.
- *φ_* is a Homomorphism.*
- $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$.
- $(\mathbb{1})_* = \mathbb{1}_*$
- *$\varphi \sim \psi$ relative to x_0 then $\varphi_* = \psi_*$.*

Proof.
- Take γ_1, γ_2 loops at x_0, then

$$\varphi(\gamma_1 \cdot \gamma_2) = \varphi(\gamma_1) \cdot \varphi(\gamma_2)$$

because

$$\varphi(\gamma_1 \cdot \gamma_2(s)) = \begin{cases} \varphi(\gamma_1(2s)) & 0 \leq s \leq 1/2 \\ \varphi(\gamma_2(2s-1)) & 1/2 \leq s \leq 1 \end{cases}$$

but

$$\varphi(\gamma_1) \cdot \varphi(\gamma_2)(s) = \begin{cases} \varphi(\gamma_1(2s)) & 0 \leq s \leq 1/2 \\ \varphi(\gamma_2(2s-1) & 1/2 \leq s \leq 1) \end{cases}$$

Therefore

$$\begin{aligned} \varphi_*([\gamma_1] \cdot [\gamma_2]) &= \varphi_*([\gamma_1 \cdot \gamma_2]) \\ &= [\varphi(\gamma_1 \cdot \gamma_2)] = [\varphi(\gamma_1) \cdot \varphi(\gamma_2)] \\ &= [\varphi(\gamma_1)] \cdot [\varphi(\gamma_2)] \\ &= \varphi_*([\gamma_1]) \cdot \varphi_*([\gamma_2]) \end{aligned}$$

- TODO
- TODO
- Let γ be a loop at x_0 and suppose that H is the homotopy between φ and ψ relative to x_0, therefore $H\colon X \times I \to Y$ s.t

  $$H(x,0) = \varphi(x), \quad H(x,1) = \psi(x), \quad H(x_0,t) = \varphi(x_0) = \psi(x_0)$$

  Then consider the map

  $$H'\colon I \times I \to Y, \quad H'(s,t) = H(\gamma(s),t)$$

  then
  $$H'(s,0) = H(\gamma(s),0) = \varphi(\gamma(s)), \quad H'(s,1) = H(\gamma(s),1) = \psi(\gamma(s))$$

  and
  $$H'(0,t) = H(x_0,t) = \varphi(x_0), \quad H'(1,t) = \psi(x_0)$$

  Therefore for any γ
  $$\varphi_*([\gamma]) = [\varphi(\gamma)] = [\psi(\gamma)] = \psi_*([\gamma])$$

draw the pictures here, it clears things up. □

Corollary 1.23. $X \cong Y \implies \pi_1(X) \cong \pi_1(Y)$.

Proof. Let $\varphi\colon X \to Y, \quad \psi\colon Y \to X$ such that

$$\psi \circ \varphi = \mathbb{1}_X, \quad \varphi \circ \psi = \mathbb{1}_Y$$

Therefore
$$(\psi \circ \varphi)_* = \psi_* \circ \varphi_* = \mathbb{1}_*$$

similarly
$$\varphi_* \circ \psi_* = \mathbb{1}_*$$

Therefore
$$\pi_1(X) \cong \pi_1(Y)$$

□

Remark 1.24. Note that the proof above would still work if we only had that $\psi \circ \varphi \sim \mathbb{1}$ relative to x_0 and $\varphi \circ \psi \sim \mathbb{1}$ relative to $\varphi(x_0)$.

Definition 1.25. Let X, Y be two spaces, we say that X is *homotopic* to Y iff there exists some $\varphi\colon X \to Y$ and $\psi\colon Y \to X$ such that

$$\psi \circ \varphi \sim \mathbb{1}_X, \quad \varphi \circ \psi \mathbb{1}_Y$$

Proposition 1.26. *Homotopy of spaces is an equivalence relation*

Proof.
- **Reflexive.** Take $\varphi = \psi = \mathbb{1}$.

- **Symmetric.** Trivial.
- **Transitive.**

Lemma 1.27. *Let $f\colon X \to Y$, $f'\colon X \to Y$ $g, g'\colon Y \to Z$ with*

$$f \sim f', \quad g \sim g'$$

then

$$g \circ f \sim g' \circ f'$$

Proof. Let H be the homotopy between f and f' and H' is the homotopy between g and g'. Then

$$H''\colon X \times I \to Z$$

defined by

$$H''(x,t) = H'(H(x,t),t)$$

notice that

$$H''(0,t) = H'(H(x,0),0) = H'(f(x),0) = g \circ f(x)$$

Similarly check the rest. □

Now suppose we have X, Y, Z spaces with

$$\varphi\colon X \to Y, \quad \varphi\colon Y \to \mathbb{Z}$$

and

$$\psi\colon Y \to X, \quad \psi' Y \to Z$$

$$\begin{tikzcd} X \arrow[r, bend right, "\varphi"'] & Y \arrow[l, bend right, "\psi"'] \arrow[r, bend right, "\varphi'"'] & Z \arrow[l, bend right, "\psi'"'] \end{tikzcd}$$

Then

$$\varphi' \circ \varphi\colon X \to Z, \quad \psi \circ \psi'\colon Z \to X$$

s.t

$$\begin{aligned} \psi \circ \psi' \circ \varphi' \circ \varphi &\sim \psi \circ \mathbb{1} \circ \varphi \\ &\sim \mathbb{1} \end{aligned}$$

□

Terminology. If $X \sim \{\cdot\}$, then we say that X is contractible.

Example 1.28. $\mathbb{R}^n$ is contractible; there is only one map $\varphi\colon \mathbb{R}^n \to \{\cdot\}$. Take the map $\psi\colon \cdot \to \mathbb{R}^n$ to be

$$\cdot \to 0$$

Indeed

$$\varphi \circ \psi \colon \cdot \to \cdot = \mathbb{1}$$

Moreover, $\psi \circ \varphi \colon \mathbb{R}^n \to \mathbb{R}^n$

$$\psi \circ \varphi(\alpha) = 0$$

let $H \colon \mathbb{R}^n \times i \to \mathbb{R}^n$ defined by

$$H(x,t) = t \cdot x$$

Similarly, we get that any convex subset of $\mathbb{R}^n$ is contractible; any star-shaped set is also contractible.

Later on we will see that S^n is not contractible.

Definition 1.29. Let X be a space and $A \subseteq X$, then $r \colon X \to A$ is called a *retraction* iff

$$r_{|A} = \mathbb{1}_{|A}$$

In this case we say that A is a *retract* of X.

Example 1.30.

- Clearly the function $r \colon \mathbb{R}^n \to \{x_0\}$ is a retraction.
- $r \colon \mathbb{R}^n \setminus \{0\} \to S^{n-1}, \quad x \to \frac{x}{|x|}$

Definition 1.31. X a space, $A \subseteq X$: we say that X deformation retracts to A (or that there is a deformation retraction from x to A) (or that A is a deformation retract of X) iff

- There is a retraction $r \colon X \to A$.
- Letting $\iota \colon A \to X$ be the inclusion map, we have

$$r \circ \iota = \mathbb{1}_{|A}$$

and

$$\iota \circ r \sim \mathbb{1}_{|X} \text{ relative to } A$$

Example 1.32.

- $\mathbb{R}^n$, convex / start shaped subsets deformation retract into a point.
- $\mathbb{R}^n$ deformation retract into D^n: let $r \colon \mathbb{R}^n \to D^n$ be defined by

$$r(x) = \begin{cases} x & \text{if } x \in D^n \\ \frac{x}{|x|} & \text{if } x \notin D^n \end{cases}$$

Let $H(x,t) = tr(x) + (1-t)x$.
- $\mathbb{R}^n \setminus \{0\}$ deformation retracts to S^{n-1}: $r \colon \mathbb{R}^n \setminus \{0\} \to S^{n-1}$ by

$$r(x) = \frac{x}{|x|}$$

$H(x,t) = (1-t)x + t\frac{x}{|x|}$ is the homotopy.
There is no deformation retraction from $\mathbb{R}^n \setminus \{0\}$ to any point.
- $I \times I$ (The closed rectangle) deformation retracts to a cup Exercise.

Proposition 1.33. *If X deformation retracts to A and A to B, then X deformation retracts to B.*

Proof. Exercise. $\square$

Theorem 1.34. $X \sim Y \implies \pi_1(X) \cong \pi_1(Y)$

Lemma 1.35. *Let X, Y topological spaces and $\varphi, \psi \colon X \to Y$ be two homotopic functions by a homotopy $H \colon X \times I \to Y$. Note that $H(x_0, \cdot)$ is a path in Y (denote it by η). Then the following diagram commutes:*

$$\begin{tikzcd} \pi_1(X, x_0) \arrow[r, "\psi_*"] \arrow[dr, "\varphi_*"'] & \pi_1(Y, \psi(x_0)) \arrow[d, "\beta_\eta"] \\ & \pi_1(Y, \varphi(x_0)) \end{tikzcd}$$

i.e

$$\beta_\eta \circ \psi_* = \varphi_*$$

Proof of lemma. Let $\gamma \colon I \to X$ be a loop based at x_0, let

$$H'(s,t) = \eta_t(s) \cdot H(\gamma(s), t) \cdot \overline{\eta_t}(s)$$

Convince yourself that the endpoints match here; therefore by the Gluing Lemma H' is continuous.

- At $t = 1$, we get
$$\eta \cdot \psi(\gamma) \cdot \overline{\eta}$$
- At $t = 0$, we get
$$\varphi(x_0) \cdot \varphi(\gamma) \cdot \overline{\varphi(x_0)}$$

So we have that

$$\begin{aligned} \eta \cdot \psi(\gamma) \cdot \overline{\eta} &\sim \varphi(x_0) \cdot \varphi(\gamma) \overline{\varphi}(x_0) \\ &\sim \varphi(\gamma) \end{aligned}$$

and therefore

$$\begin{aligned} [\eta \cdot \psi(\gamma) \cdot \overline{\eta}] &= [\varphi(\gamma)] \\ \beta_\eta[\psi(\gamma)] &= \varphi_*([\gamma]) \\ \beta_\eta(\psi_*([\gamma])) &= \varphi_*([\gamma]) \end{aligned}$$

therefore

$$\varphi_* = \beta_\eta \circ \psi_*$$

$\square$

Proof of the theorem.

$$(Y, y_0) \xrightarrow{\quad \psi \quad} (X, x_0) \xrightarrow{\quad \varphi \quad} (Y, y_1) \xrightarrow{\quad \psi \quad} (X, x_1)$$

we know that

$$\psi \circ \varphi \sim \mathbb{1}, \quad \varphi \circ \psi \sim \mathbb{1}$$

Therefore by the lemma

$$(\psi \circ \varphi)_* = \beta, \quad (\varphi \circ \psi)_* = \beta'$$

But β, β' are isomorphisms, therefore we get that

$$\varphi_* \text{ is injective and surjective}$$

□

1.4 Fundamental group of all spheres

Theorem 1.36.

$$\pi_1(S^n) = \begin{cases} e & \textit{if } n \geq 2 \\ \mathbb{Z} & \textit{if } n = 1 \end{cases}$$

Proof.
- $\pi_1(S^2)$; there is a homeomorphism between $S^2 \setminus \{N\}$ and $\mathbb{R}^2$ (N is the North pole) called the stereographic projection. i.e $t \to (0,0,1) + t(x,y,z) = (tx, ty, tx+1)$ since $1 + tz = 0 \implies t = \frac{t}{1-z}$ we get the map Σ given by

  $$\Sigma(x,y,z) = (\frac{x}{1-z}, \frac{y}{1-z})$$

  For the inverse map $t \to (0,0,1) + t(\alpha, \beta, -1) = (t\alpha, t\beta, 1-t)$; on the sphere

  $$t^2\alpha^2 + t^2\beta^2 + (1-t)^2 = 1 \implies t = \frac{2}{\alpha^2 + \beta^2 + 1}$$

  so its inverse is given by

  $$\Sigma^{-1}(\frac{2}{\alpha^2+\beta^2+1}, \frac{2\beta}{\alpha^2+\beta^2+1}, 1 - \frac{2}{\alpha^2+\beta^2+1})$$

  TODO, do the same for $\mathbb{R}^n$.
  We shall take our loops to be based at the south pole S. Let $\gamma\colon I \to S^2$ be such a loop. If it happens that γ never passes through N; $\Sigma \circ \gamma$ is a loop in $\mathbb{R}^2$ based at 0. Thus $H\colon I \times I \to S^2$ given by

  $$H(s,t) = \Sigma^{-1}\left((1-t)\Sigma \circ \gamma(s)\right)$$

  gives a homotopy between γ and the constant loop at the south pole.
  Now suppose that we have any loop $\gamma\colon I \to S^2$. Consider the preimage by γ of the open Northern hemisphere, this is open in I since γ is continuous; in fact it is open in $(0,1)$ (and hence an open subset of $\mathbb{R}$). Now any open subset of $\mathbb{R}$ is the union of countably many disjoint open intervals. Consider now the preimage of N, it must be closed in $[0,1]$ and hence must be compact. Therefore since the preimage of the open Northern hemisphere contains that of N; the preimage of N is an open cover of the preimage of N. So it has a finite subcover. Thus there are finitely many disjoint open intervals which contain the

preimage of N. Consider one of these intervals (a,b). Note that $\gamma((a,b)) \subseteq N$; since γ is continuous we can conclude that $\gamma(a), \gamma(b) \in \overline{N}$ (the closed Northern hemisphere). Note that $\gamma(a), \gamma(b)$ must be on the equator; since if they were in the Northern hemisphere we would have that a belongs to one of the other *disjoint* intervals to (a,b). Note that the closed Northern hemisphere is homeomorphic to D^2 by $(x,y,z) \to (x,y)$. but note that $\partial D^2 = S^1$ which is path connected; since we can find some path $\eta \colon I \to S^1$ going from $\gamma(a) \to \gamma(b)$ with η homotopic $\gamma_{|[a,b]}$.
Now we can write a homotopy between γ and some loop that skips N; see the picture we are using the homotopy above on all (a,b) like above. This is continuous by the gluing lemma according to our homotopy. So γ is homotopic to a loop that skips N and so homotopic to the constant loop.

- We will think of S^1 as
$$\{z \in \mathbb{C} \colon |z| = 1\}$$
We will base our loops at the point $z = 1$. We define
$$\Phi \colon \mathbb{Z} \to \pi_1(S^1), \quad \Phi(n) = [e^{2\pi ins}]$$
We also define
$$p \colon \mathbb{R} \to S^1, \quad p(x) = e^{2\pi ix}$$
Note that $[p(ns)] = \Phi(n)$. We will show that Φ is a homomorphism
$$\begin{aligned}\Phi(n) \cdot \Phi(m) &= [p(ns)] \cdot [p(ms)] \\ &= [p(ns) \cdot p(ms)] \\ &= [p(ns) \cdot p(n + ms)] \\ &= [p(ns \cdot (n + ms))]\end{aligned}$$
Note that $ns \cdot (n + ms)$ is a path $0 \to m + n$, but $\mathbb{R}$ is simply connected, hence
$$ns \cdot (n + ms) \sim (n+m)s \implies p(ns \cdot (n+ms)) \sim p((n+m) \cdot s)$$
Therefore
$$\Phi(n) \cdot \Phi(m) = [p(n+m)s] = \Phi(n+m)$$
Claim 1. For any path γ in S^1 starting at 1, there is a unique path $\tilde{\gamma} \in \mathbb{R}$ starting at 0, s.t
$$p(\tilde{\gamma}) = \gamma$$
we say that $\tilde{\gamma}$ is the *lift* of γ starting at 0.
Claim 2. Suppose that γ_1, γ_2 are two paths in S^1 starting at 1, which are homotopic. Then their lifts starting at 0 are also homotopic.

Φ **is surjective.** Let $[\gamma] \in \pi_1(S^1)$, then we know that there is a unique $\tilde{\gamma}$ in $\mathbb{R}$ s.t

$$\tilde{\gamma}(0) = 0, \quad p(\tilde{\gamma}) = \gamma$$

Especially $p(\tilde{\gamma}(1)) = \gamma(1) = 1$, hence

$$\tilde{\gamma}(1) = n \in \mathbb{Z} \implies \tilde{\gamma} \sim ns$$

Therefore

$$p(\tilde{\gamma}) \sim p(ns) \implies \gamma \sim p(ns) \implies [\gamma] = [p(ns)] = \Phi(n)$$

Φ **is injective.** Suppose that $\Phi(n) = \Phi(m)$ therefore

$$[p(ns)] = [p(ms)]$$

We have that the lift of $p(ns)$ is ns and the lift of $p(ms)$ is ms but

$$p(ns) \sim p(ms) \implies ns \sim ms \implies n = m$$

We will prove the two claims in the following setting: $\tilde{X}, X$ two spaces, $p\colon \tilde{X} \to X$ s.t $\forall p \in X,\ \exists U \ni p$ open then $p^{-1}(U)$ is a disjoint union of open sets s.t each one of them (say V) satisfies

$$p_{|V} \quad \text{is a homeomorphism onto } U$$

Such a U is called an elementary neighborhood or an evenly covered neighborhood. p is called the projection covering map and

$$(\tilde{X}, p) \text{ is called a covering space of } X$$

The example to take in mind here is $X = S^1$, $\tilde{X} = \mathbb{R}$ with p "wrapping" $\mathbb{R}$ around S^1. Indeed consider $U_1 = S^1 \setminus \{1\}$, then $p^{-1}(U_1)\ \mathbb{R} \setminus \mathbb{Z}$; and $U_2 = S^1 \smallsetminus \{-1\}$ then $p^{-1}(U_2)$ and $U_2 = \mathbb{R} \setminus (1/2\mathbb{Z})$, so $(\mathbb{R}, p)$ is indeed a covering space of S^1.

Claim 3. Suppose that $(\tilde{X}, p)$ is a covering space of X, and let $f\colon Y \times I \to X$ and suppose we have some $\tilde{f}\colon Y \times \{0\} \to \tilde{X}$ s.t

$$p \circ \tilde{f} = f$$

then $\tilde{f}$ can be extended *uniquely* to $Y \times I$ s.t $p \circ \tilde{f} = f$ (draw the picture).

Proof that claim 3 implies claim 1 and 2. For claim 1, let $Y = \{\cdot\}$, in this case

$$\{\cdot\} \times I \cong I$$

so we have that $f\colon I \to X$ and $\tilde{f}\colon 0 \to \tilde{X}$ s.t $\tilde{f}$ can be uniquely extended to I; which is precisely claim 1 where we are given a path in X and the starting point of the lift is $\tilde{X}$. For claim 2, take $Y = I$ and f to be the homotopy between the two paths in X; from claim 1 we can lift the homotopy at the initial time; claim 3 lifts the homotopy to $\tilde{X}$. By uniqueness of lifts the sides of the homotopy lift to the constant paths and the top of the

homotopy gives a lift of the second path starting where the lift of the first path started.
Proof of claim 3. Let $y \in Y$, $\forall t \in I$ there is an open neighborhood O_t of t in I and an open neighborhood N_t of y in Y s.t

$$f(N_t \times O_t) \subseteq \text{ an elementary neighborhood}$$

$\{y\} \times I$ is compact and $\{N_t \times O_t\}_{t \in I}$ is an open cover, therefore $\exists t_1, \cdots, t_k \in I$ s.t

$$\{y\} \times I \subseteq (N_{t_1} \times O_{t_1}) \cup \cdots \cup (N_{t_k} \times O_{t_k})$$

Let $N = \bigcap_{i=1}^{k} N_{t_i}$, and let $\delta > 0$ be the Lebesgue number of the cover $O_{t_1}, \cdots, O_{t_k}$. Therefore any closed interval of length $\leq \delta$ is contained in one of the O_t's. Consider $N \times [0, \delta]$ we know that $f(N \times [0, \delta]) \subset U$ an elementary neighborhood. But $p \circ \tilde{f} = f$, therefore $\tilde{f}(N \times 0) \subset p^{-1}(U)$, let $\tilde{U}$ be the "part" of $p^{-1}(U)$ which contains $\tilde{f}(y, 0)$. Now by taking the preimage of $\tilde{U}$ by $\tilde{f}$ we get some open subset of Y which is also an open subset of N; replace N with this smaller subset call it N. Now p is a homeomorphism between $\tilde{U}$ and U, so it has an inverse $p^{-1} \colon U \to \tilde{U}$, define

$$\tilde{f} = p^{-1} \circ f$$

This way, we have extended $\tilde{f}$ to $N \times [0, \delta]$. Now do the same for $N \times [\delta, 2\delta]$ by looking at $\tilde{f}(N \times \{\delta\})$ and making sure that it maps to only one of the preimages of the new elementary neighborhood, we shrink N again and extend $\tilde{f}$ by letting it be $p^{-1} \circ f$.
Note that by construction we have that $p \circ \tilde{f} = f$, and it is continuous by the Gluing lemma. Hence we know that $\forall y \in Y$, $\exists N_y$ s.t $\tilde{f}$ is extended continuously to include $N_y \times I$ with $p \circ \tilde{f} = f$. Let us show that lifts of paths are unique; indeed suppose that we have $\gamma \colon I \to X$ a path in X, and let $\tilde{\gamma}_1,\ \tilde{\gamma}_2$ be two lifts of γ (i.e $p \circ \tilde{\gamma}_1 = p\tilde{\gamma}_2 = \gamma$ and they both have the same start point). Therefore $\exists \delta > 0$ s.t any closed interval of length δ is mapped by γ into an elementary neighborhood. Consider the interval $[0, \delta]$, we know that

$$\gamma\left([0, \delta]\right) \subset \text{ an elementary neighborhood } U$$

This implies that $\tilde{\gamma}_1([0, \delta]) \subset p^{-1}(U)$ and $\tilde{\gamma}_2([0, \delta]) \subset p^{-1}(U)$; since $[0, \delta]$ is connected, we know that

$$\gamma_1([0, \delta]),\ \gamma_2([0, \delta]) \subset \text{ a single } \tilde{U}_1, \tilde{U}_2$$

since $\tilde{\gamma}_1(0) = \tilde{\gamma}_2(0)$ we get $\tilde{U}_1 = \tilde{U}_2$. But since $p \circ \tilde{\gamma}_1 = p \circ \tilde{\gamma}_2$ and since p is a homeomorphism, then

$$\tilde{U}_1 = \tilde{U}_2 \to 0$$

Proceeding similarly, we get that $\tilde{\gamma}_1 = \tilde{\gamma}_2$.
Note that restricting f to a line like $\{y\} \times I$ gives a path in X, and restricting $\tilde{f}$ to this line gives us a lift of this path. So if we use $\tilde{f}$'s coming from different $N_y \times I$'s, then they must coincide on the intersection because the intersection is made of lines as above. So $\tilde{f}$ and $\tilde{f}$ is unique. And $\tilde{f}$ is continuous using the gluing lemma for open sets.

□

Theorem 1.37 (Fundamental theorem of Algebra). *Let* $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ *then*

$$p(z_0) = 0 \textit{ for some } z_0 \in \mathbb{C}$$

Proof. Assume that $p(z) \neq 0$ consider

$$H_1(s,t) = \frac{\frac{p(tre^{2\pi is})}{|p(tre^{2\pi is}|)}}{\frac{p(tr)}{|p(tr)|}}$$

Note that this gives a homotopy between the constant loop in S^1 at 1 and $\frac{\frac{p(re^{2\pi is})}{|p(re^{2\pi is})|}}{\frac{p(r)}{|r|}}$. Choose $r > 0$ to be s.t $r > 1$ and $r > |a_{n-1}| + \cdots + |a_0|$, then notice that if z is restricted to the circle of radius r, we have that

$$\begin{aligned}
|z|^n = r^n = r^{n-1} \cdot r &> r^{n-1}(|a_{n-1}| + \cdots + |a_0|) \\
&> r^{n-1}|a_{n-1}| + \cdots + r^{n-1}|a_0| \\
&> r^{n-1}|a_{n-1}| + r^{n-2}|a_{n-2}| + \cdots + |a_0| \\
&= |a_{n-1}z^{n-1}| + \cdots + |a_0| \\
&\geq |a_{n-1}z^{n-1} + \cdots + a_0|
\end{aligned}$$

it follows that the following polynomials depending on t never vanish on the circle of radius r:

$$q_t(z) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$$

This is because for $|z| = r$ we have that

$$q_t(z) \geq |z^n| - |t(a_{n-1}z^{n-1} + \cdots + a_0)| > 0$$

Now consider $H_2(s,t)$ to be

$$H_2(s,t) = \frac{\frac{q_t(re^{2\pi is})}{|q_t(re^{2\pi is}|)}}{\frac{q_t(r)}{|p(r)|}}$$

Note that this gives a homotopy between the loop which we just created (write it out TODO); and

$$\frac{\frac{r^n e^{2\pi ins}}{r^n}}{r^n/r^n} = e^{2\pi ns}$$

Therefore using both H_1 and H_2 we have a homotopy between the constant loop and $e^{2\pi ins}$ by contradiction. □

Theorem 1.38 (Brouwer's fixed point). $f\colon D^n \to D^n$ *then* f *has a fixed point, i.e* $\exists p_0 \in D^n$ *s.t*

$$f(p_0) = p_0$$

Proof.

- **Case** $n = 1$. Let $f\colon D^1 \to D^1$, if $f(-1) = -1$ we are done same if $f(1) = 1$. Otherwise, we get that $f(-1) > -1,\ f(1) < 1$ therefore $f(x) - x$ is > 0 at -1 and < 0 at 1. Therefore by the IVT, $\exists x_0 \in (-1, 1)$ s.t

$$f(x_0) - x_0 = 0$$

- **Case** $n = 2$. Assume that $f\colon D^2 \to D^2$ s.t $f(x) \neq x$ for all $x \in D^2$. Define $g\colon D^2 \to S^1$ as follows: for every $x \in D^2$, take the line from $f(x)$ to x then $g(x)$ is the intersection between this line and S^1. Note that g is continuous and it is also a retraction from $D^2 \to S^1$. Take the loop $e^{2\pi is}$ in D^2, then $e^{2\pi is} \sim$ constant in D^2. But

$$g(e^{2\pi is}) = e^{2\pi is} \not\sim \text{ constant loop in } S^1$$

Contradiction.
Another way of saying this is that

$$g_*\colon \pi_1(D^2) \to \pi(S^1)$$

but $\pi_1(D^2)$ is trivial, but its image is not the identity.

□

Corollary 1.39. *There is no retraction $D^2 \to S^1$.*

Theorem 1.40 (Borsuk-Ulam Theorem)**.** *Let $f\colon S^n \to \mathbb{R}^n$ then there $\exists \in S^n$ s.t*

$$f(x) = f(-x)$$

Proof.

- $n = 1$. We have $f\colon S^1 \to \mathbb{R}$, if $f(1) = f(-1)$ we are done. Otherwise wlog assume $f(1) > f(-1)$; consider

$$g(x) = f(x) - f(-x)$$

then g is continuous with $g(1) > 0$ and $g(-1) < 0$ therefore by the IVT applied to the upper hemisphere, $\exists x_0 \in S^1 \cap \{(x, y) \mid y \geq 0\}$ s.t $g(x_0) = 0$.
- $n = 2$. Suppose that $f\colon S^2 \to \mathbb{R}^2$ with $f(x) \neq f(-x)$ for all x, let $g\colon S^2 \to S^1$ be defined by

$$g(x) = \frac{f(x) - f(-x)}{|f(x) - f(-x)|}$$

note that $g(-x) = -g(x)$. Also wlog we are going to assume that

$$g(1, 0, 0) = 1$$

Consider now the loop

$$\gamma'\colon I \to S^2, \quad \gamma'(s) = (\cos(2\pi s), \sin(2\pi s), 0)$$

Let $\gamma\colon I \to S^1$ be $g \circ \gamma'$ we know that $\pi^1(S^2) = e$ therefore

$$\gamma' \sim \text{ constant}$$

and hence

$$\gamma \sim \text{ constant in } S^1$$

Note that $\gamma'(s + 1/2) = -\gamma'(s)$ so it follows that $\gamma(s + 1/2) = -\gamma(s)$. Consider the lift $\tilde{\gamma}$ of γ starting at 0. Then (do math), we get that

$$\tilde{\gamma}(s + 1/2) - \tilde{\gamma}(s) = \frac{n(s)}{2}$$

Where $n(s)$ is odd; but since LHS is continuous $n(s)$ must be a constant n. i.e

$$\tilde{\gamma}(s + 1/2) = \tilde{\gamma}(s) + n/2$$

Hence

$$\tilde{\gamma}(1) = \tilde{\gamma}(1/2) + n/2 = \tilde{\gamma}(0) + n = n$$

Since γ is homotopic to the constant, $\gamma(1) = 0$ but n is odd: contradiction.

□

1.5 First Algebraic Detour

Let $\{G_\alpha\}_{\alpha \in \mathcal{A}}$ be a collection of groups, we can make a group out of the product in two ways

1. $\bigtimes_{\alpha \in \mathcal{A}} G_\alpha$ is the set of all elements in the set theoretic product with multiplication defined pointwise.

2. **The weak product.** $\pi_{\alpha \in \mathcal{A}} G_\alpha$ is the subgroup of the product s.t for each element only finitely many components are nontrivial.

Theorem 1.41 (Universal Property of the Weak Product of Abelian Groups)**.** *Suppose that we have a collection $\{G_\alpha\}$ all abelian, then $\prod_\alpha G_\alpha$ satisfies the following property: Suppose we have homomorphisms $f_\alpha\colon G_\alpha \to A$ where A is abelian then $\exists! f\colon \prod_\alpha G_\alpha \to A$ a homeomorphism s.t the following diagram commutes for every α:*

$$\begin{tikzcd} G_\alpha \arrow[r, "\iota_\alpha"] \arrow[dr, "f_\alpha"'] & \prod_\beta G_\beta \arrow[d, "f"] \\ & A \end{tikzcd}$$

Where

$$\iota_\alpha(g) = \{g_\beta\} = \begin{cases} g & \text{if } \alpha = p \\ e & \text{if } \alpha \neq p \end{cases}$$

Proof. Let $f(\{g_\alpha\}) = \prod_\alpha f_\alpha(g_\alpha)$. Note that this product is well defined since only finitely many $f_\alpha(g_\alpha)$'s are nontrivial. Moreover, the order is irrelevant since A is Abelian. Clearly we have that f satisfies the diagram. We also have

$$f(\{g_\alpha\}\{h_\alpha\}) = f(\{g_\alpha h_\alpha\}) = \prod_\alpha f_\alpha(g_\alpha h_\alpha) = \prod f_\alpha(g_\alpha) \prod f(h_\alpha)$$

Where the last step is justified since A is abelian.

Note that $\prod_\alpha G_\alpha$ is generated by the $i_\alpha(G_\alpha)$'s, but note that any f' that satisfies the diagram must agree with our f on each of these G_α's (both give us $f_\alpha(G_\alpha)$) therefore $f' = f$. □

Remark 1.42. Note that the universal property of the weak product defines it (up to isomorphism). Indeed, suppose that we have a collection $\{G_\alpha\}$ all abelian, and some abelian G that satisfies the following property: For any A and any homomorphisms $f_\alpha \colon G_\alpha \to A$ and homomorphisms $\iota_\alpha \colon G_\alpha \to G$ where A is abelian and $\exists! f \colon G \to A$ a homomorphism s.t the following diagram commutes for every α:

$$\begin{tikzcd} G_\alpha \arrow[r, "\iota_\alpha"] \arrow[dr, "f_\alpha"'] & G \arrow[d, "f"] \\ & A \end{tikzcd}$$

Then $G \cong \prod_{\alpha \in \mathcal{A}} G_\alpha$.

Proof. Draw the commutative diagram. □

Definition 1.43. Let S be a set. The *free abelian group on S* is an abelian group $F(S)$ together with a map

$$\iota \colon S \to F(S)$$

such that for every abelian group A and every map $\chi \colon S \to A$ there exists a unique group homomorphism

$$\varphi \colon F(S) \to A$$

making the following diagram commute:

$$\begin{tikzcd} S \arrow[r, "\iota"] \arrow[dr, "\chi"'] & F(S) \arrow[d, "\varphi"] \\ & A \end{tikzcd}$$

Example 1.44. Let $S = \{x\}$, let $F(S) = \mathbb{Z}$ and $j \colon \{x\} \to \mathbb{Z}$ be defined by $j(x) = 1$. Pick any A abelian, let $\chi \colon S \to A$ let $a = \chi(x)$ an define

$$\varphi \colon \mathbb{Z} \to A, \quad \varphi(n) = a^n$$

indeed, $\varphi(j(x)) = a = \chi(x)$ abd φ is unique since it $\varphi'(1) = a$ and 1 generates $\mathbb{Z}$.

Generally we write elements of $F(\{x\})$ as nx.

Theorem 1.45. *The free abelian group of S is unique up to isomorphism.*

Proof. Draw the picture; final element is unique. □

Theorem 1.46. *Suppose that $S = \bigcup_{\alpha \in \mathcal{A}} S_\alpha$ with $\alpha \neq \alpha' \implies S_\alpha \cap S'_\alpha = \varnothing$, then*

$$F(S) \cong \prod_\alpha F(S_\alpha)$$

Proof.

$$\begin{tikzcd}
S_\alpha \arrow[r, "j_{S_\alpha}"] \arrow[d, hook, "i_\alpha"'] & F(S_\alpha) \arrow[r, "\iota_\alpha"] \arrow[d, "\varphi_\alpha"'] & \prod_{\beta \in \mathcal{A}} F(S_\beta) \arrow[d, dashed, "\Phi"] \\
S \arrow[r, "j_S"'] \arrow[rr, bend right, "\chi"'] & F(S) \arrow[r, "\overline{\chi}"'] & A
\end{tikzcd}$$

□

Corollary 1.47. *For any S, note that*

$$F(S) \cong \prod_{x \in S} F(\{x\}) \cong \prod_{x \in S} \mathbb{Z}$$

We write elements in $F(S)$ as

$$n_1 x_1 + \cdots + n_k x_k$$

This notation is useful because multiplication in $F(S)$ corresponds to addition of coefficients.

Theorem 1.48. *Any abelian group A is the homomorphic image of a free abelian group.*

Proof. Let $S = A$ and let $\chi \colon S \to A = \mathbb{1}$; draw the diagram the proof is done. □

Example 1.49. Let $\mathbb{Z}_2$, take $A = \mathbb{Z}_2$ and find the $\varphi \colon F(A) \to A$. But also let $\varphi \colon \mathbb{Z} \to \mathbb{Z}_2$ defined by

$$\varphi(n) = n \mod 2$$

Definition 1.50. let A be an abelian group suppose that $F(S)$ is free abelian on S and

$$\varphi \colon F(S) \to A \quad \text{surjective}$$

Every nontrivial element of the kernel of φ is said to be a non-trivial relation on the generators of A. Moreover if $r_1, \cdots, r_k$ are nontrivial relation and $r_{k+1} \in \langle r_1, \cdots, r_k \rangle$, then we say that r_{k+1} is a consequence of $r_1, \cdots, r_k$. If $r_1, \cdots, r_k$ generate the whole kernel, we say that we have a presentation of A in terms of generators and relations.

Example 1.51. In the example above write $\mathbb{Z} = F(\{x\})$, think of A as being generated by x with the following relations imposed:

$$x + x = 0, \quad x + x + x + x = 0, \quad -x - x = 0, \cdots$$

Note that $x + x + x + x$ is a consequence of $x + x$. Indeed, since all relations are a consequence of $x + x = 0$, therefore $\mathbb{Z}_2$ is generated by a single element with a single relation.

Free Groups

Definition 1.52 (Free group)**.** Let S be a set. A *free group* on S, denoted $F^*(S)$ (often written $F(S)$), is a group together with a function

$$j : S \to F^*(S)$$

such that for every group G and every function

$$\chi : S \to G$$

there exists a unique group homomorphism

$$\varphi : F^*(S) \to G$$

making the following diagram commute:

$$\begin{tikzcd} S \arrow[r, "j"] \arrow[dr, "\chi"'] & F^*(S) \arrow[d, "\varphi"] \\ & G \end{tikzcd}$$

i.e. $\varphi \circ j = \chi$.

Equivalently, $(F^*(S), j)$ is the initial object in the category of groups equipped with a map from S.

Theorem 1.53 (Uniqueness)**.** *The free group on S is unique up to isomorphism.*

Proof. Let (F_1, j_1) and (F_2, j_2) be two free groups on S. By the universal property, there exist unique homomorphisms

$$\phi : F_1 \to F_2, \qquad \psi : F_2 \to F_1$$

such that $\phi \circ j_1 = j_2$ and $\psi \circ j_2 = j_1$. Then $\psi \circ \phi : F_1 \to F_1$ is a homomorphism satisfying $(\psi \circ \phi) \circ j_1 = j_1$. By uniqueness it must be the identity, hence $\psi \circ \phi = \mathrm{id}$. Similarly $\phi \circ \psi = \mathrm{id}$, so $F_1 \cong F_2$. □

Example 1.54. If $S = \{x\}$ then $F^*(S) \cong \mathbb{Z}$.

Proof. Define $j(x) = 1 \in \mathbb{Z}$. Given any group G and map $\chi : S \to G$, write $\chi(x) = g$. Define

$$\varphi : \mathbb{Z} \to G, \qquad \varphi(n) = g^n.$$

This is the unique homomorphism with $\varphi \circ j = \chi$, hence $\mathbb{Z}$ satisfies the universal mapping property. □

Free Products

Definition 1.55 (Free product)**.** Let $\{G_\alpha\}_{\alpha\in\mathcal{A}}$ be a family of groups. The *free product*

$$\prod_{\alpha\in\mathcal{A}}^{*} G_\alpha$$

is a group together with homomorphisms

$$i_\alpha : G_\alpha \to \prod_{\alpha}^{*} G_\alpha$$

such that for any group G and any family of homomorphisms

$$f_\alpha : G_\alpha \to G$$

there exists a unique homomorphism

$$f : \prod_{\alpha}^{*} G_\alpha \to G$$

making the diagrams commute:

$$\begin{tikzcd} G_\alpha \arrow[r, "i_\alpha"] \arrow[dr, "f_\alpha"'] & \prod_\alpha^* G_\alpha \arrow[d, "f"] \\ & G \end{tikzcd}$$

i.e. $f \circ i_\alpha = f_\alpha$ for all α.

Free Groups as Free Products

Theorem 1.56. *Suppose $S = \bigsqcup_\alpha S_\alpha$ is a disjoint union of sets. Then*

$$F^*(S) \cong \prod_{\alpha}^{*} F^*(S_\alpha).$$

Proof. Both sides satisfy the same universal mapping property for maps from S. Hence they are isomorphic by uniqueness of universal objects. □

Corollary 1.57. *Let S be any set. Then*

$$F^*(S) \cong \prod_{s\in S}^{*} F^*(\{s\}) \cong \prod_{s\in S}^{*} \mathbb{Z}.$$

Corollary 1.58. *Every group is a homomorphic image of a free group.*

Proof. Let G be a group and take $S = G$ as a set. The identity map $S \to G$ extends uniquely to a homomorphism

$$F^*(S) \to G$$

which is surjective. □

Generators and Relations

Definition 1.59. Let G be a group and suppose

$$f : F^*(S) \twoheadrightarrow G$$

is a surjective homomorphism.

- The elements of S (via their images in G) are called generators.
- The nontrivial elements of $\ker(f)$ are called relations.
- If $r_1, \dots, r_k \in F^*(S)$ then r_{k+1} is a consequence of them if

$$r_{k+1} \in \langle\langle r_1, \dots, r_k \rangle\rangle,$$

  the smallest normal subgroup containing $r_1, \dots, r_k$.

Existence of Free Products

Theorem 1.60. *Free products exist.*

Construction via reduced words. Let $\mathcal{A} = \bigsqcup_\alpha G_\alpha$ be the disjoint union of the underlying sets.

Words. A word is a finite sequence

$$w = g_1 g_2 \cdots g_n \quad \text{with } g_i \in G_{\alpha_i}.$$

A word is *reduced* if:

1. $g_i \neq e$ for all i,
2. adjacent letters come from different groups: $\alpha_i \neq \alpha_{i+1}$.

The empty word is denoted by e.

Multiplication. Define multiplication by:

- concatenation of words,
- followed by reduction:
  - multiply adjacent elements from the same group,
  - remove identities,
  - cancel inverse pairs.

Group axioms.

- Identity: the empty word.
- Inverse:

$$(g_1 \cdots g_n)^{-1} = g_n^{-1} \cdots g_1^{-1}.$$

- **Associativity.** Let W denote the set of reduced words. For each letter g (i.e. an element of some G_α), define the map

$$L_g : W \to W, \qquad L_g(w) = g \cdot w,$$

where the product on the right means concatenation followed by reduction.
Let $w = h_1 \cdots h_n$ be a reduced word. Then

$$L_{g^{-1}} \circ L_g(h_1 \cdots h_n) = L_{g^{-1}}(gh_1 \cdots h_n).$$

We consider cases:
  - If g and h_1 belong to different groups, no reduction occurs, and

$$L_{g^{-1}}(gh_1 \cdots h_n) = h_1 \cdots h_n.$$

  - If g and h_1 belong to the same group, two subcases arise:
    * If $h_1 \neq g^{-1}$, then multiplication inside the group replaces gh_1 by a single element, and applying $L_{g^{-1}}$ restores $h_1 \cdots h_n$ after reduction.
    * If $h_1 = g^{-1}$, cancellation occurs and

$$L_{g^{-1}}(gh_1 \cdots h_n) = g^{-1}h_2 \cdots h_n = h_1h_2 \cdots h_n.$$

Hence $L_{g^{-1}} \circ L_g = \mathrm{id}$, so L_g is a bijection with inverse $L_{g^{-1}}$.
Now let $w = g_1 \cdots g_n$ be a reduced word and define

$$L_w := L_{g_1} \circ \cdots \circ L_{g_n}.$$

Then L_w is a bijection as a composition of bijections.
Moreover, the map

$$w \longmapsto L_w$$

is injective. Indeed, if $w_1 \neq w_2$, then

$$L_{w_1}(e) = w_1 \neq w_2 = L_{w_2}(e),$$

where e denotes the empty word.
Suppose

$$w_1 = g_1 \cdots g_n h_1 \cdots h_m, \qquad w_2 = h_m^{-1} \cdots h_1^{-1} k_1 \cdots k_\ell,$$

so that reduction gives

$$w_1 \cdot w_2 = g_1 \cdots g_n k_1 \cdots k_\ell.$$

Then

$$L_{w_1 \cdot w_2} = L_{g_1} \circ \cdots \circ L_{g_n} \circ L_{k_1} \circ \cdots \circ L_{k_\ell}.$$

On the other hand,

$$L_{w_1} \circ L_{w_2} = L_{g_1} \circ \cdots \circ L_{g_n} \circ L_{h_1} \circ \cdots \circ L_{h_m} \circ L_{h_m^{-1}} \circ \cdots \circ L_{h_1^{-1}} \circ L_{k_1} \circ \cdots \circ L_{k_\ell},$$

and the middle terms cancel since $L_{h_i} \circ L_{h_i^{-1}} = \mathrm{id}$. Thus

$$L_{w_1} \circ L_{w_2} = L_{w_1 \cdot w_2}.$$

Finally,

$$L_{(w_1 \cdot w_2) \cdot w_3} = (L_{w_1} \circ L_{w_2}) \circ L_{w_3} = L_{w_1} \circ (L_{w_2} \circ L_{w_3}) = L_{w_1 \cdot (w_2 \cdot w_3)}.$$

Since the map $w \mapsto L_w$ is injective, we conclude

$$(w_1 \cdot w_2) \cdot w_3 = w_1 \cdot (w_2 \cdot w_3),$$

establishing associativity.

We still need to show that this group respects the UMP of the Free product. Indeed, we define

$$i_\alpha \colon G_\alpha \to W, \quad i_\alpha(g) = g$$

This is a valid homomorphism since

$$i_\alpha(g \cdot h) = (gh) = g \cdot h = i_\alpha(g) \cdot i_\alpha(h)$$

Suppose that we have G and

$$f_\alpha \colon G_\alpha \to G$$

Let $f \colon w \to G$ be defined as follows:

$$f(g_1 \cdots g_n) = f_{\alpha_1}(g_1) f_{\alpha_2}(g_2) \cdots f_{\alpha_n}(g_n)$$

Note that f does make the diagram commute, take $g \in G_\alpha$ we get

$$f_\alpha(g) = f(i_\alpha(g))$$

We still need to show that it is a homomorphism, let $w = g_1 \cdots g_n h_1 \cdots h_m$ and $w_2 = h_m^{-1} \cdots h_1^{-1} k_1 \cdots k_l$ s.t $w_1 \cdot w_2 = g_1 \cdots g_n k_1 \cdots k_l$ then

$$\begin{aligned} f(w_1) f(w_2) &= f_{\alpha_1}(g_1) \cdots f_{\alpha_n}(g_n) f_{\beta_1}(h_1) \cdots f_{\beta_m}(h_m) f_{\beta_m}(h_m^{-1}) \cdots f_{\beta_1}(h_1) f_{\gamma_1}(k_1) \cdots f_{\gamma_l}(k_l) \\ &= f(w_1 \cdot w_2) \end{aligned}$$

Note that any homomorphism that makes the diagram commute must agree with f on the one letter words, which generate the product. Therefore f is indeed unique. $\square$

2 The Van Kampen Theorem

Let X be a space with

$$X = \bigcup_{\alpha \in \mathcal{A}} A_\alpha, \quad \bigcap_{\alpha \in \mathcal{A}} A_\alpha \neq \varnothing$$

Fix $x_0 \in \bigcap_{\alpha \in \mathcal{A}} A_\alpha$, therefore all the fundamental groups below will be based at x_0. Let $\iota_\alpha \colon A_\alpha \to X$ be the usual inclusion map. So we get

$$\iota_\alpha^* \colon \pi_1(A_\alpha) \to \pi_1(X)$$

Thus we get a unique homomorphism

$$\Phi \colon \prod_{\alpha \in \mathcal{A}}^{*} \pi_1(A_\alpha) \to \pi_1(X)$$

such that the following diagram commutes:

$$\begin{tikzcd}
\pi_1(A_\alpha, x_0) \arrow[r, "i_\alpha"] \arrow[dr, "\iota_\alpha"'] & \prod\limits_{\beta \in \mathcal{A}}^{*} \pi_1(A_\beta, x_0) \arrow[d, "\Phi"] \\
& \pi_1(X, x_0)
\end{tikzcd}$$

Theorem 2.1 (The Van Kampen Theorem). *Suppose we have the setting above, and A_α's are open, $A_\alpha \cap A_\beta$ is path connected, $A_\alpha \cap A_\beta \cap A_\gamma$ is path connected. Then $\Phi \colon \prod_\alpha^* \pi_1(A_\alpha) \to \pi_1(X)$ is surjective and* $\ker \Phi$ *is the smallest normal subgroup generated by elements of the form*

$$\left[i_\alpha\left(\iota_{\alpha\beta}^*([\gamma])\right)\right]^{-1}\left[i_\beta\left(\iota_{\beta\alpha}^*([\gamma])\right)\right]$$

Where

$$[\gamma] \in \pi_1(A_\alpha \cap A_\beta)$$

and

$$\iota_{\alpha\beta} \colon A_\alpha \cap A_\beta \to A_\alpha \quad \textit{inclusion}$$

Remark 2.2. Note that

$$\begin{aligned}
\Phi\left[i_\alpha\left(\iota_{\alpha\beta}^*([\gamma])\right)\right] &= \iota_\beta^*\left(\iota_{\beta\alpha}^*([\gamma])\right) \\
&= (\iota_\beta \circ \iota_{\beta\alpha})^*(\gamma)
\end{aligned}$$

Similarly

$$\begin{aligned}
\Phi\left[i_\beta\left(\iota_{\beta\alpha}^*([\gamma])\right)\right] &= \iota_\alpha^*(\iota_{\alpha\beta}^*([\gamma])) \\
&= (\iota_\alpha \circ \iota_{\alpha\beta})^*[\gamma]
\end{aligned}$$

Since $\iota_\beta \circ \iota_{\beta\alpha} = \iota_\alpha \circ \iota_{\alpha\beta}$ we have

$$(\iota_\beta \circ \iota_{\beta\alpha})^* = (\iota_\alpha \circ \iota_{\alpha\beta})^*$$

So a priori we have that if H Is the smallest normal subgroup generated by these, we have that (in general)

$$H \subset \ker \Phi$$

The theorem gives us

$$\ker \phi = H$$

We will now demonstrate that all the conditions above are necessary:

Example 2.3 (Double intersection)**.** Take S_1 as the open union of two arcs (more than half-arcs). Note that their intersection is not path connected. Well

$$\pi_1(S_1) = \mathbb{Z},\ \pi_1(P_1) = \pi_1(P_2) = e$$

Example 2.4 (Triple intersection)**.** View picture, lots of drawing.

Example 2.5 (A_α open)**.** Hawaiian earrings; i.e take circles of radius $1/n$ centered at $(1/n, 0)$. Let A_α's be the circles, if SVK applies, then the fundamental group would be

$$\prod_{\text{circles}}^{*} \mathbb{Z}$$

This would imply that any loop in this space is homotopic to a loop contained in finitely many circles. However, it is easy to see that this is not the case (we will show this in an assignment).

Proof. To show surjectivity, we need to show that for any loop $\gamma \in X$ we can find finitely many loops $\gamma_1, \cdots, \gamma_n$ each one in some A_{α_i}. Such that $\gamma \sim \gamma_1 \cdots \gamma_n$. Since the A_α's are open, we have that $\gamma^{-1}(A_\alpha)$ is open in I. Therefore

$$\left\{\gamma^{-1}(A_\alpha)\right\}_{\alpha\in\mathcal{A}} \text{ is an open cover of } I$$

Let $\delta > 0$ be a Lebesgue number of this cover, i.e any closed interval of length δ is contained in some $\gamma^{-1}(A_\alpha)$. i.e for any closed interval of length δ, its image by γ is contained in some A_α. We are going to use the fact that if the domain of a path is decomposed into intervals, then the original path is homotopic to the product of the paths obtained by restricting the original path to these intervals. In our case, we get that $\gamma \sim \gamma_1 \cdot \gamma_2 \cdots \gamma_n$, where each γ_i is contained in some A_{α_i}. Let $x_i = \gamma_i(1)$, since $A_\alpha \cap A_\beta$ is path connected, we can take some path η_i from $x_0 \to x_i$ contained in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$. Notice that

$$\gamma \sim (\gamma_1 \cdot \overline{\eta}_1) \cdot (\eta_1 \cdot \gamma_2 \cdot \overline{\eta_2}) \cdots (\gamma_{n-1} \cdot \gamma_n)$$

Where the $(\bullet)$ are loops in the A_α's. Now by combining adjacent loops which live in the same A_α's and removing those that are nullhomotopic in A_α, we get what we want.

To show that the kernel of Φ is N (the normal subgroup described in the theorem), it is enough to show that

$$\Phi(w_1) = \Phi(w_2) \implies w_2^{-1} w_1 \in N$$

Let $w_1 \in K$ and $w_2 = e$ therefore

$$\Phi(e) = \Phi(w_1)$$

hence $ew_1 \in N$.

Given a loop γ in X, a factorization of γ is a not necessarily reduced word from the alphabet $\bigcup_{\alpha \in \mathcal{A}} \pi_1(A_\alpha)$, i.e a finite string of the form

$$[\gamma_1][\gamma_2]\cdots[\gamma_n], \quad [\gamma_i] \in \pi_1(A_{\alpha_i})$$

such that

$$\gamma \sim \gamma_1 \cdot \gamma_2 \cdots \gamma_n$$

by the surjectivity above, we get that every loop has a factorization.

Moreover, we say that two factorization of any loop are equivalent if they can be related by a finite sequence of the following operations:

1. Adding or removing identities.
2. combining or splitting adjacent letters if they come from the same group.
3. $$[\gamma_1]\cdots[\gamma_i]_\alpha\cdots[\gamma_n] \rightleftarrows [\gamma_1]\cdots[\gamma_i]_\beta\cdots[\gamma_n]$$
   where if $\gamma_i \in A_\alpha \cap A_\beta$, we replace $\gamma_i \in A_\alpha$ to its inclusion to A_β. i.e we are changing
   $$\iota^*_{\alpha\beta}[\gamma_i] \rightleftarrows \iota^*_{\beta\alpha}[\gamma_i]$$

Claim. Two factorizations are equivalent $\implies$ the corresponding reduces words are in the same coset of N.

Proof of claim. Moves 1 and 2 do not change the reduced words. Consider two words that are related by move 3, therefore

$$\begin{aligned}[\gamma_n]^{-1}\cdots[\gamma_i]_\alpha^{-1}\cdots[\gamma_1]^{-1}[\gamma_1]\cdots[\gamma_i]_\beta\cdots[\gamma_n] &= [\gamma_n]^{-1}\cdots[\gamma_i]_\alpha^{-1}[\gamma_i]_\beta^{-1}\cdots[\gamma_n] \\ &= (\cdots[\gamma_n])^{-1}[\gamma_i]_\alpha^{-1}[\gamma_i]_\beta(\cdots[\gamma_n]) \in N\end{aligned}$$

(Since $[\gamma_i]_\alpha^{-1}[\gamma_i]_\beta$ is a generator of N, by definition).

So suppose that we have $w_1 = [\gamma_1]\cdots[\gamma_n],\ w_2 = [\gamma_1']\cdots[\gamma_m']$ s.t $\Phi(w_1) = \Phi(w_2)$; i.e

$$[\gamma_1]\cdots[\gamma_n] \sim [\gamma_1']\cdots[\gamma_m']$$

So $[\gamma_1]\cdots[\gamma_n]$ and $[\gamma_1']\cdots[\gamma_m']$ are two factorization of the same loop. Since

$$\gamma_1\cdots\gamma_n \sim \gamma_1'\cdots\gamma_m'$$

we have a homotopy

$$H \colon I \times I \to X$$

Note that $\{H^{-1}(A_\alpha)\}_\alpha$ is an open cover of $I \times I$, therefore $\exists \delta > 0$ s.t any closed square of sides $\leq 2\delta$ is contained in one of these those sets. i.e its image by H is contained in one of the A_α's. Take $I \times I$, divide it into squares of size $\leq \delta \times \delta$ wlog we can assume that the endpoints of the γ's are in the grid already.

By construction associate to each of the rectangles and A_α to which it goes by H. By moving the vertical edges of every other row, we can make sure that at no point is shared by more than three rectangles, note that not to move the first and last make sure that we have odd number of rows. Now for every corner of every rectangle, we take a path from x_0 to the image of that corner by H in the intersection of the A_α's sharing that corner. For the points on the lowest and highest edges, also require that these paths are in A_α's in which the corresponding γ_i' lives.

Note that any line that goes from left to right, gives a loop by restricting H to this line. Note that we can associate to this loop several factorizations by inserting $\overline{\eta}\eta$ at every corner. View the picture.

Claim. All these factorizations are equivalent and are equivalent to $\gamma_1 \cdots \gamma_n$ and $\gamma'_1 \cdots \gamma'_m$.
Subclaim 1. All factorizations corresponding to the same line are equivalent. This is clear since the different choices of A_α's correspond exactly to move 3.
Subclaim 2. The factorizations corresponding to the bottom / top edges are equivalent to $\gamma_1 \cdots \gamma_n$ / $\gamma'_1 \cdots \gamma'_n$. By applying move 3 at every edge if needed. We make sure that the edges making up γ_i are in the A_α corresponding to it. Then apply move 2 to combine the loops into γ's (or γ's).
Subclaim 3. Two lines that differ by a square (picture blue and red) give equivalent factorizations. By subclaim 1, we pick the factorizations that coincide everywhere away from the square; Moreover, for those on the square we choose the A_α which correspond to that rectangle. However, the loops corresponding to these two are homotopic. This is because the image of two homotopic paths under a continuous function are homotopic.

This proves the claim, and hence the theorem. □

Corollary 2.6. *$\pi_1(S^n) = e$ for $n \geq 2$.*

Proof. Cut the sphere into two 2/3-spheres. Note that both 2/3-spheres (S_1, S_2) are homeomorphic to a disk (which is nullhomotopic), therefore By VK

$$\pi_1(S) = \frac{\pi_1(S_1) \star \pi_1(S_2)}{N} \cong e$$

□

Definition 2.7. Let (X_1, x_1) with $x_1 \in X_1$ and (X_2, x_2) with $x_2 \in X_2$ s.t $X_1 \cap X_2 = \varnothing$, we define the wedge sum these two pointed spaces to be the space $X_1 \vee X_2$ to be the quotient space $X_1 \cup X_2/(x_1 \sim x_2)$.

Note that this is easily generalizable to $X_1 \vee X_2 \vee \cdots \vee X_n$, or even more arbitrary collection.

Example 2.8. $S^{n_1} \vee S^{n_n} \vee \cdots \vee S^{n_k}$ is called a bouquet of spheres.

Corollary 2.9. $\pi_1(S^{n_1} \vee \cdots \vee S^{n_k}) \cong \mathbb{Z} \star \cdots \mathbb{Z}$, *number of* S^1 *times.*

Proof. Divide the bouquet into S^{n_i}'s $+\varepsilon$ (view pic), notice that the double intersections are just the "nibbles", same for the triple intersections. Therefore by SVK

$$\begin{aligned}\pi_1(S^{n_1} \vee \cdots \vee S^{n_k}) &= \pi_1(S^{n_1}) \star \cdots \star \pi_1(S^{n_k}) \\ &= \prod_{S^{n_i}=S^1} \mathbb{Z}\end{aligned}$$

□

2.1 Cell complexes.

Definition 2.10. A cell complex is a space constructed in the following way:

- X_0 is a finite discrete space (finitely many points).
- Having constructed X_n, construct X_{n+1} by taking finitely many (disjoint) D^{n+1}. For each one, take a continuous function $\partial D^{n+1} \cong S^n \to X^n$. Define X^{n+1} to be the quotient of $X^n \cup (\text{Disks})/[x \sim \text{ its image}]$.

We will always stop (in this course) at some finite n, then $X = X_n$.

We call X_i the i'th squeleton of X.

$X_{i+1} - X_i$ is a union of open D^{i+1} which are called $i+1$-cells.

Example 2.11. View pictures.

Example 2.12. A Klein bottle is two Mobius strips glued along their boundary circles.

The projective plane is D^2 glued with a Mobius strip along its boundary S^1.

Example 2.13. Double holed torus; triple holed torus.

Theorem 2.14. *Let X be a topological space, suppose Y is obtained from X by attaching a collection of n-cells. Then*

$$\pi_1(Y) \cong \begin{cases} \pi_1(X) & \text{if } n \geq 3 \\ \pi_1(X)/N & \text{if } n = 2 \end{cases}$$

Where N is the smallest normal subgroup generated by the loops

$$\gamma_i \varphi_i(\partial D^2)\overline{\gamma_i}$$

Where γ_i is a path from the base of the fundamental group (x_0) to the image of 1 by the φ_i.

(View the picture, this makes sense).

Proof. We are going to consider the space Z obtained from Y as in the picture. Clearly, Z deformation retracts to Y, hence $r_*,\ i_*$ are isomorphisms so $\pi_1(Z, z_0) \cong \pi_1(Y, x_0)$. Let $A\&B$ be the following sets,

$$A := Z - X, \quad B = Z - \{\text{centers of the } n\text{-cells}\}$$

Clearly, $A\&B$ are open and $A \cap B$ is path-connected and nonempty since $z_0 \in A \cap B$. Therefore by SVK

$$\pi_1(Z) \cong \frac{\pi_1(A) \star \pi_1(B)}{N}$$

Notice that $A \sim \cdot \implies \pi_1(A) = e$. OTOH B deformation retracts to X, hence

$$\pi_1(B, z_0) = \pi_1(X, x_0)$$

Moreover, note that this deformation retraction induces a retraction which sends the "extra stuff" in B to $\gamma_1\varphi(S^1)$.

N is the smallest normal subgroup generated by injecting loops in $A \cap B$ into B (since A is nullhomotopic). Note that $A \cap B$ is homotopic to the wedge of lollipops, SVK gives us that $\pi_1(A \cap B) = \pi_1(\ell_1) \star \pi_1(\ell_2)$, using β_η (view picture) we can easily compute $\pi_1(\ell_1),\ \pi_1(\ell_2)$. Therefore

$$\pi_1(A \cap B) \text{ is generated by the obvious two obvious loops}$$

Therefore looking only at the generated, we know that N is the only normal subgroup generated by these two loops. Therefore

$$\begin{aligned}
\pi_1(B, z_0)/N &= \frac{\pi_1(B, z_0)}{[\ell_1, \ell_2]} \\
&\cong \frac{\pi_1(X, x_0)}{[r(\ell_1), r(\ell_2)]} \\
&\cong \frac{\pi_1(X, x_0)}{[\gamma_i\varphi_i(S^1)\overline{\gamma_i}]}
\end{aligned}$$

□

Example 2.15 (Fundamental group of the g-holed torus)**.** By the usual construction of the torus, we can have

$$\begin{aligned}
\pi_1(T_1) &= \frac{\langle a \rangle \star \langle b \rangle}{[aba^{-1}b^{-1}]} \\
&= \mathbb{Z} \times \mathbb{Z}
\end{aligned}$$

For T_2 we have

$$\pi_1(T_2) = \frac{a \star b \star c \star d}{[aba^{-1}bcdc^{-1}d^{-1}]}$$

Corollary 2.16. *Suppose G is a group $\implies \exists X$ with $\pi_1(X) \cong G$.*

Proof. Any G can be written a quotient of a free group, let X be the bouquet of S^1 on generators. Glue a 2-cell for each relation. □

3 Covering Spaces.

Reminder. We have $\tilde{X} \xrightarrow{p} X$ such that $p^{-1}(U)$ is a union of a collection of disjoint open sets s.t p is a homeomorphism from each one to U. U is called an elementary neighborhood.

Some examples: [insert table from screenshot]

We have already proven the following proposition:

Proposition 3.1. *Suppose we have $\tilde{X} \xrightarrow{p} X$ and $f \colon Y \times I \to X$, then we can extend and $\tilde{f} \colon Y \times \{0\} \to \tilde{X}$ s.t $p \circ \tilde{f} = f$, then we can extend $\tilde{f}$ uniquely to $Y \times I$ s.t*

$$p \circ \tilde{f} = f$$

Corollary 3.2. *γ path in X starting at x_0 and let $\tilde{x_0}$ be s.t*

$$p(\tilde{x_0}) = x_0$$

Then there is a unique $\tilde{\gamma}$ starting at $\tilde{x_0}$ s.t

$$p(\tilde{\gamma}) = \gamma$$

Then $\tilde{\gamma}$ is called the lift of γ starting at $\tilde{x_0}$.

Corollary 3.3. *If $\gamma_1,\ \gamma_2$ in X and $\gamma_1 \sim \gamma_2$ then $\tilde{\gamma_1} \sim \tilde{\gamma_2}$ (both lifts start at the same spot).*

Lemma 3.4.

1. $\tilde{\overline{\gamma}} = \overline{\tilde{\gamma}}$.
2. $\gamma_1 \tilde{\cdot} \gamma_2 = \tilde{\gamma_1} \cdot \tilde{\gamma_2}$

Note that this is implicitly assuming that we are picking endpoints in a way that matches. In other words, for $\tilde{\overline{\gamma}}$ we pick the start point to be the end point of $\tilde{\gamma}$. Similarly, we pick the start point of $\tilde{\gamma_2}$ to be the endpoint of the endpoint of $\tilde{\gamma_1}$, moreover we need $\gamma_1 \tilde{\cdot} \gamma_2$ to have the same startpoint as the lift of $\tilde{\gamma_1}$.

Proof.

1. $p(\overline{\tilde{\gamma}}) = \overline{\gamma}$ and $p(\tilde{\overline{\gamma}}) = \overline{\gamma}$, since they have the same starting point, by uniqueness of lifts we get that these two paths are the same.
2. Clearly we have that $p(\gamma_1 \tilde{\cdot} \gamma_2) = \gamma_1 \cdot \gamma_2$, OTOH we have that $p(\tilde{\gamma_1} \cdot \tilde{\gamma_2}) = \gamma_1 \cdot \gamma_2$. Uniqueness of lift gives us the that they coincide.

□

Theorem 3.5. *Suppose that we have $\tilde{X} \xrightarrow{p} X$, hence*

$$p_* \colon \pi_1(\tilde{X}, \tilde{x_0}) \to \pi_1(X, x_0)$$

then

1. *p_* is injective.*
2. *$\mathrm{Im}(p_*)$ consists of all loops which loops which lift to loops.*

Proof. Suppose that $p_*[\gamma_1] = p_*[\gamma_2] \implies [p(\gamma_1)] = [p(\gamma_2)]$. Note that γ_1 and γ_2 are lifts of $p(\gamma_1)$ and $p(\gamma_2)$. Hence

$$\gamma_1 \sim \gamma_2 \iff [\gamma_1] = [\gamma_2]$$

It remains to show that $p_*(\pi_1(\tilde{X}, \tilde{x_0})) =$ equivalence classes of loops that lift to loops. Indeed pick $p_*[\gamma] \in \pi_1(\tilde{X}, \tilde{x_0})$, hence

$$p_*[\gamma] = [p(\gamma)]$$

But $p(\gamma)$ is a loop and its lift is γ which is a loop. OTOH let γ be some loops s.t. its lift $\tilde{\gamma}$ is also a loop, therefore

$$p(\tilde{\gamma}) = \gamma \implies [\gamma] \in p_*(\pi_1(\tilde{X}, \tilde{x_0}))$$

□

Definition 3.6. Suppose we a $\tilde{X} \xrightarrow{p} X$ covering space, pick any $x \in X$ then $|p^{-1}|$ is called the number of sheets of the covering space.

Proposition 3.7. *The number of sheets of a covering space is well defined.*

Proof. Suppose that we have $x, y \in X$, fix some γ a path $x \to y$. Define the map

$$\varphi \colon p^{-1}(x) \to p^{-1}(y), \quad \varphi(\tilde{x}) = \tilde{\gamma}(1)$$

Where $\tilde{\gamma}$ is the lift of γ starting at $\tilde{x}$. Similarly define

$$\psi \colon p^{-1}(y) \to p^{-1}(x), \quad \psi(\tilde{y}) = \tilde{\overline{\gamma}}(1)$$

Where the lift starts at $\tilde{y}$. If φ and ψ are mutually inverses, then they are bijections and we are done. Indeed, this stems from the fact that $\tilde{\overline{\gamma}} = \overline{\tilde{\gamma}}$. □

Theorem 3.8. *The number of sheets of the covering space is equal to the number of cosets of $p_*(\pi(\tilde{X}, \tilde{x_0}))$ is $\pi(X, x_0)$.*

Proof. Let $G = p_*(\pi_1(\tilde{X}, \tilde{x_0})) \subset \pi_1(X, x_0)$, define

$$\varphi \colon \text{Cosets} \to p^{-1}(x_0), \quad \varphi(G[\gamma]) = \tilde{\gamma}(1)$$

where $\tilde{\gamma}$ is the lift of γ starting at $\tilde{x_0}$.

- **Well defined.** Suppose that $G[\gamma_1] = G[\gamma_2]$, therefore $\exists g \in G$ with

  $$\gamma_2 = g\gamma_1$$

  but therefore $\tilde{\gamma_1}$ and $\tilde{\gamma_2}$ differ by $\tilde{g}$ on the right, which is a loop (by the proposition above). Hence their endpoint is the same.
- **surjection.** Easy, take any path $\gamma \colon \tilde{x_0} \to \tilde{x}$ with $p(\tilde{x}) = x_0$, therefore $p(\gamma)$ is a loop at x_0.
- **Injection.** Suppose that we have two γ_1, γ_2 s.t $\tilde{\gamma_1}(1) = \tilde{\gamma_2}(1)$, then

  $$\tilde{\gamma_1} \cdot \overline{\tilde{\gamma_2}} \text{ is a loop in } \tilde{X}$$

  but

  $$\tilde{\gamma_1} \cdot \overline{\tilde{\gamma_2}} = \gamma_1 \tilde{\cdot} \overline{\gamma_2}$$

  but then this means that $\gamma_1 \cdot \overline{\gamma_2}$ is a loop that lifts to a loop, therefore

  $$G[\gamma_1] = G[\gamma_2]$$

□

Definition 3.9. A space X is said to be locally path connected iff for any $x \in X$ and any $U \ni x$ (open neighborhood), then $\exists V \ni x$ an open neighborhood which is path connected and $x \in V \subset U$

Theorem 3.10 (The Lifting Criterion). *Let $(\tilde{X}, p)$ be a covering space of X, fix $x_0 \in X$, $\tilde{x_0} \in \tilde{X}$ (with $p(\tilde{x_0}) = x_0$) and take $Y \xrightarrow{f} X$ with $f(y_0) = x_0$. Suppose that Y is path connected and locally path connected, therefore $\exists!\ \tilde{f}\colon Y \to \tilde{X}$ such that $p \circ \tilde{f} = f \iff f_*(\pi_1(Y, y_0)) \subset p_*(\pi(\tilde{X}, \tilde{x_0}))$*

Proof. $\implies$ Suppose that we have we have such a $\tilde{f}$, therefore

$$(p \circ \tilde{f})_*(\pi_1(Y, y_0)) = f_*(\pi(Y, y_0))$$

Notice that the RHS is

$$p_*(\tilde{f}_*(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x_0}))$$

Hence

$$f_*(\pi(Y, y_0)) \subset p_*(\pi_1(\tilde{X}, \tilde{x_0}))$$

As desired.

$\impliedby$ Define $\tilde{f}$ as follows, for any $y \in Y$ take γ to be a path from $y_0 \to y$ and take the image of this path in X by f and then lift this image up to $\tilde{X}$ getting a path starting at $\tilde{x_0}$; the endpoint of this lifted path is $\tilde{f}(y)$. Note that $p \circ \tilde{f}(y) = f(x)$ by construction.

Assuming that $\tilde{f}$ is continuous and well defined, suppose that we have two lifts of f (namely $\tilde{f_1},\ \tilde{f_2}$), pick any point $y \in Y$ and take a path $\gamma\colon\ y_0 \to y$, then note that $\tilde{f_1}(\gamma),\ \tilde{f_2}(\gamma)$ are both lifts of the same path $f(\gamma)$ starting at $\tilde{x_0}$ Therefore by uniqueness of lifts of paths we get that $\tilde{f_1}(\gamma) = \tilde{f_2}(\gamma)$ in particular they have the same endpoint.

We now show well definedness. Suppose that γ' is another path $y_0 \to y$, notice that

$$f(\gamma) \cdot \overline{f(\gamma')} = f(\gamma \cdot \overline{\gamma'})$$

Since $f_*(\pi_1(Y, y_0)) \subset p^*(\pi_1(\tilde{X}, \tilde{x_0}))$, we get that $f(\gamma \cdot \gamma')$ is a loop which lifts to a loop based at $\tilde{x_0}$. This means that if you take a lift of $f(\gamma)$ starting at x_0 and follow it by the lift of $f(\overline{\gamma'})$ starting at the end of the lift of $f(\gamma)$ we will finish at $\tilde{x_0}$. Therefore it follows that if $f(\gamma')$ starting at $\tilde{x_0}$ we end up at the endpoint of $f(\gamma)$, therefore $\tilde{f}$ is well defined.

Let O be an open subset of $\tilde{x}$ and let $y \in \tilde{f}^{-1}(O)$ let U be an elementary neighborhood containing $f(y)$ and $\tilde{U}$ the part of the preimage of U which contains $\tilde{f}(y)$. Note that $\tilde{U} \cap O$ is open hence $p(\tilde{U} \cap O)$ is open (in U which is open hence in X). But f is continuous, therefore $f^{-1}(p(O \cap \tilde{U})) \ni y$ is open (in Y). Since Y is locally path connected, $\exists V \ni y$ an open neighborhood which is path connected s.t. $V \subset f^{-1}(p(O \cap \tilde{U}))$. Let γ be a path $y_0 \to y$, then $f(\gamma)$ is a path in X and $\widetilde{f(\gamma)}$ is the lift of $f(\gamma)$ starting at $\tilde{x_0}$; let $y' \in V$ since V is path connected we can take some $\eta\colon y \to y'$ s.t. $\eta \subset V$. We are going to use the path $\gamma \cdot \eta$ to find $\tilde{f}(y')$. Consider $f(\gamma \cdot \eta) = f(\gamma) \cdot f(\eta)$ and take its lift starting at $\tilde{x_0}$: $\widetilde{f(\gamma \cdot \eta)} = \widetilde{f(\gamma)} \cdot \widetilde{f(y)}$. But $\tilde{f}(\eta) = p^{-1} \circ f(\eta)$ (uniqueness of lifts) note that this means that

$\tilde{f}(y') = \tilde{f}(\eta)(1) \in U \cap O \subset O$ therefore

$$\tilde{f}(V) \subset O$$

$\square$

Theorem 3.11. *Let X be path connected and locally path connected and locally simply connected, then there is a covering space $(\tilde{X}, p)$ s.t. $\tilde{X}$ is simply connected.*

In this case we call $\tilde{X}$ the universal covering space of X.

Proof. Pick $x_0 \in X$, let $\tilde{X} = \{[\gamma] : \gamma$ is a path in X starting at $x_0\}$, let $p \colon \tilde{X} \to X$ be defined by

$$p([\gamma]) = \gamma(1)$$

Consider the set of all simply connected neighborhoods in X. Define the following collection of sets on $\tilde{X}$:

$$U_{[\gamma]} = \{\gamma \cdot \eta \colon \eta \text{ a path in } U,\ \gamma(0) = x_0,\ \gamma(1) \in U,\ U \text{ is simply connected}\}$$

Suppose that $[\gamma'] \in U_{[\gamma]}$ we claim that

$$U_{[\gamma]} = U_{[\gamma']}$$

Indeed, in this case we have that $\gamma' = \gamma \cdot \eta$ for some $\eta \subset U$. Note that

$$U_{[\gamma']} = \{[\gamma' \cdot \eta']\} = \{[\gamma \cdot \eta \cdot \eta']\} \subset U_{[\gamma]}$$

But also

$$U_{[\gamma]} = \{[\gamma \cdot \eta']\} = \{\gamma \cdot \eta \cdot \overline{\eta} \cdot \eta'\} = \{\gamma' \cdot \overline{\eta} \cdot \gamma'\} \subset U[\gamma']$$

We want to show that $\left\{U_{[\gamma]}\right\}_{\gamma,U}$ is a valid basis for a topology. Indeed, Consider $U[\gamma],\ V[\gamma']$ and suppose that $U \cap V \neq \varnothing$, let $[\gamma'']$ belong to this intersection. This implies that $U_{[\gamma]} = U_{[\gamma'']}$ and $V_{[\gamma']} = V_{[\gamma'']}$ hence

$$\gamma''(1) \in U \cap V$$

Therefore by local simply connectedness, there is some simply connected W s.t. $\gamma''(1) \in W$ and $W \subset U \cap V$. Now consider $W_{[\gamma'']} = \{[\gamma'' \cdot \eta] \colon \eta \in W\}$ but $W \subset U$ and $W \subset V$ therefore

$$W_{[\gamma'']} \subset \{[\gamma'' \cdot \eta] \colon \eta \in U\} = U, \quad W_{[\gamma'']} \subset V$$

as desired. Let $\tau\left(\left\{U_{[\gamma]}\right\}_{U,\gamma}\right)$ be the topology on $\tilde{X}$. We show that the elementary neighborhoods are the simply connected neighborhoods in X, indeed, let U be such a neighborhood in X. Consider $p^{-1}(U)$, note that

$$p^{-1}(U) = \{\text{set of all equivalence classes of paths which end in } U\} = \bigcup_{[\gamma]} U_{[\gamma]}$$

check that the second equality above holds. Suppose that $U_{[\gamma]} \cap U_{[\gamma']} \neq \varnothing$, i.e. $[\gamma''] \in U_{[\gamma]} \cap U_{[\gamma']} \implies U_{[\gamma]} = U_{[\gamma'']} = U_{[\gamma']}$. This means that $p^{-1}(U)$ is a collection of disjoint open sets; some collection of $U_{[\gamma]}$'s. Fix such a $U_{[\gamma]}$, note that $p_{|U_{[\gamma]}}$ is a bijection with U. Surjectivity follows from the fact that for any point in U, we can take η inside U and goes from $\gamma(1)$ this point. Now suppose that $p([\gamma \cdot \eta]) = p([\gamma \cdot \eta'])$, therefore η and η' have the same endpoint and they must have the same starting point (endpoint of γ), but U is simply connected therefore $[\eta] = [\eta']$. Note that p is an open function; is open. To see this it is enough to show that $p(U_{[\gamma]})$ is open for any $U_{[\gamma]}$; but notice that $p(U_{[\gamma]}) = U$ and hence p is open. To see that p is continuous; notice that the U's form a basis for X with $p^{-1}(U) = \bigcup U_{[\gamma]}$ which is open.

We want to show that $\tilde{X}$ is path-connected, we will do so by showing that for every $[\gamma]$ in $\tilde{X}$, there is a path from $[x_0]$ to $[\gamma]$. Let $\Gamma\colon I \to [\tilde{X}]$ be given by

$$\Gamma(t) = [\gamma_t]$$

Notice that

$$p(\Gamma(t)) = p[\gamma_t] = \gamma_t(1) = \gamma(t)$$

Also note that

$$\Gamma(0) = [x_0], \quad \Gamma(1) = [\gamma]$$

It remains to show that Γ is continuous, consider $\Gamma^{-1}(U_{[\gamma']})$ and let $t_0 \in \Gamma^{-1}(U_{[\gamma']})$. Hence $\Gamma(t_0) \in U_{[\gamma']}$, therefore

$$[\gamma_{t_0}] \in U_{[\gamma']} \implies U_{[\gamma']} = U_{[\gamma_{t_0}]}$$

Hence

$$\gamma_{t_0}(1) \in U \implies \gamma_{t_0} \in U$$

Since γ is continuous we have that $\exists \delta > 0$ s.t.

$$\gamma(t_0 - \delta, t_0 + \delta) \subset U$$

Then if $t \in (t_0 - \delta, t_0 + \delta)$ we get $\gamma(t) \in U$, therefore if $t \in (t_0 - \delta, t_0 + \delta)$ we have

$$\Gamma(t) \in U_{\gamma_{t_0}}$$

This holds because $\Gamma(t) = [\gamma_t] = [\gamma_{t_0} \cdot \gamma_{t_0 t}]$. Since $\gamma_{t_0 t} \subset U$ it follows that

$$[\gamma_t] \in U_{[\gamma_{t_0}]}$$

We now show that $\tilde{X}$ is simply connected, we do this by showing that $\operatorname{Im} p_* = \{e\}$, this is enough since p_* is injective. So consider a loop γ which lifts to a loop $\Gamma = \Gamma(t) = [\gamma_t]$ notice that this being a loop we get

$$\Gamma(0) = \Gamma(1) \implies [x_0] = [\gamma]$$

Hence γ is homotopic to the constant loop and hence $\tilde{X}$ is simply connected. □

Theorem 3.12. *Let X be a path connected, locally path connected and locally simply connected space;*

Let G be a subgroup of $\pi_1(X, x_0)$. Then there is a covering space $(\tilde{X}_G, p_G)$ of X s.t.

$$(p_G)_*(\pi(\tilde{X}_G, \tilde{x})) = G$$

Proof. Construct $\tilde{X}$ as before. Define the following relation on $\tilde{X}$

$$[\gamma_1] \sim [\gamma_2] \iff [\gamma_1] \cdot [\overline{\gamma}_2] = [\gamma \cdot \overline{\gamma_2}] \in G$$

This is an equivalence relation:

- **Reflexive:** $[\gamma \cdot \overline{\gamma}] = [x_0] \in G$
- **Symmetric:** $[\gamma_1 \cdot \overline{\gamma}_2] = [\overline{\gamma_1} \cdot \gamma_2]^{-1} \in G$ but G closed under inverses.
- **Transitive:** Suppose $[\gamma_1 \cdot \overline{\gamma_2}] \in G$ and $[\gamma_2 \cdot \overline{\gamma}_3] \in G$ but

$$\begin{aligned}
[\gamma_1 \cdot \gamma_3] &= [\gamma_1 \cdot x_0 \cdot \overline{\gamma}_3] \\
&= [\gamma_1 \cdot \overline{\gamma_2} \cdot \gamma_2 \cdot \overline{\gamma}_3] \in G
\end{aligned}$$

Let

$$\tilde{X}_G = \tilde{X} / \sim$$

and let $q\colon \tilde{X_G} \to \tilde{X}_G$ be the quotient map;

$$q([\gamma]) = [[\gamma]]$$

We thus get immediately that $\tilde{X}_G$ is a path-connected topological space.

Notice that if $[\gamma_1] \sim [\gamma_2]$ then

$$p([\gamma_1]) = p([\gamma_2])$$

Therefore p is constant on the equivalence classes, hence it descends to a continuous map to $\tilde{X}_G$ by

$$p_G([[\gamma]]) = p_G([\gamma])$$

Notice that p_G is an open map. Indeed let $O \subset \tilde{X}_G$ open therefore $q^{-1}(O)$ is open in $\tilde{X}$. But $p(q^{-1}(O)) = p_G(O)$ is open since p is open. Take $U[\gamma]$ and $U_{[\gamma']}$ and suppose we have $[\eta] \in U_{[\gamma]}$ and $[\eta'] \in U_{[\gamma']}$ are equivalent. Note that $U_{[\eta]} = U_{[\gamma]}$ and $U_{[\eta']} = U_{[\gamma']}$. We are going to assume wlog that $[\gamma] \sim [\gamma']$. Pick any point in U and suppose that $[\gamma \cdot \eta] \in U_{[\gamma]}$ and $p(\gamma \cdot \eta) = x$ and $[\gamma' \cdot \eta'] \in U_{[\gamma]}$ and $p(\gamma' \cdot \eta') = x$. Now look at $[\gamma \cdot \eta\overline{(\gamma' \cdot \eta')}] = [\gamma \cdot \eta \cdot \overline{\eta} \cdot \overline{\gamma'}]$ but U is simply connected, therefore

$$[\gamma \cdot \eta\overline{(\gamma' \cdot \eta')}] = [\gamma \cdot \overline{\gamma'}] \in G$$

Therefore

$$[\gamma \cdot \eta] \sim [\gamma' \sim \eta']$$

Notice that this implies that $q^{-1}(q(U_{[\gamma]}))$ is a collection of $U_{[\gamma]}$'s. Pick any point, let U be an elementary neighborhood in $\tilde{X}$, we claim that it is also an elementary neighborhood for $\tilde{X}_G$. To see this, take the preimage by p_G, consider $p_G^{-1}(U) =$ all equivalence classes of $\tilde{X}$ which project by p to U. But this is also the set of all equivalence classes of $U_{[\gamma]}$'s which is equal to a disjoint union of subsets of $U_{[\gamma]}$'s each one of them is open since $U_{[\gamma]}$'s are open. Restrict P_G to $q(U_{[\gamma]})$, it is obvious that p_G is onto, because p is onto. Similarly p_G is injective since all the elements in $q(U_{[\gamma]})$ have different endpoints. There $(\tilde{X}_G, p_G)$ is a covering space for X.

Note that we can also show that $\tilde{X}$ is a covering space for $\tilde{X}_G$. Exercise.

Consider γ a loop in X at x_0 whose lift to $\tilde{X}_G$ starting at $[[x_0]]$ is a loop. It is easy to see that this lift $q(\Gamma)$ where Γ is the lift to $\tilde{X}$ starting at $[x_0]$. This is because

$$p_G \circ q(\gamma) = p(\Gamma) = \gamma$$

with

$$q(\Gamma(0)) = q(\Gamma(1)) \iff [[x_0]] = [[\gamma]] \iff [\gamma \cdot \overline{x_0}] \in G \iff [\gamma] \in G$$

Therefore the set of loops that lift to loops is exactly G. □

Definition 3.13. Let $(\tilde{X}_1, p_1)$ and $(\tilde{X}_2, p_2)$ be covering spaces of X, we say that $(\tilde{X}_1, p_1)$ is isomorphic to $(\tilde{X}_2, p_2)$ if there is a homeomorphism $f\colon \tilde{X}_1 \to \tilde{X}_2$ s.t.

$$p_1 = p_2 \circ f$$

Theorem 3.14. *Suppose that X is locally path-connected, suppose $(\tilde{X}_1, p_1)$ and $(\tilde{X}_2, p_2)$ are two covering space and take $\tilde{x}_1, \tilde{x}_2$ in their respective spaces. Then the covering spaces are isomorphic by a unique isomorphism mapping $\tilde{x}_1 \to \tilde{x}_2 \iff (p_1)_*(\pi_1(\tilde{X}_1, \tilde{x_2})) = (p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2))$*

Proof. $\implies$ Notice that

$$(p_1)_*(\pi_1(\tilde{X}, \tilde{x_1})) = (p_2)_*(f_*(\pi_1(\tilde{X}_1, \tilde{x}_1))) = (p_2)_*(\pi_1(\tilde{X}_2, \tilde{x}_2))$$

$\impliedby$ Note that since X is locally path connected, we have that $\tilde{X}_1$ is locally path connected, therefore by the lifting criterion we have some $\tilde{p}_1\colon \tilde{X}_1 \to \tilde{X}_2$ and similarly $\tilde{p}_2\colon \tilde{X}_2 \to \tilde{X}_1$. By uniqueness of the lift $\tilde{X}_1 \to \tilde{X}_1$ we have that

$$\tilde{p}_2 \circ \tilde{p}_1 = \mathbb{1}$$

Similarly for $\tilde{p}_1 \circ \tilde{p}_2$.

□

Theorem 3.15. *There is a $1-1$ correspondence between subgroups of $\pi_1(X, x_0)$ and isomorphism classes of covering space of X with a selected point above x_0. If you want to ignore this selected point, then there is a $1-1$ correspondence between conjugacy classes of subgroups of $\pi_1(X, x_0)$ and isomorphism equivalence classes of covering spaces.*

Proof. The bijection is defined by

$$[(\tilde{X}, p, \tilde{x_0})] \rightleftarrows p_*(\pi_1(\tilde{X}, \tilde{x_0}))$$

Now that this map is well defined by the last theorem. This map is onto by the existence of a covering space s.t. $p_*(\pi_1(\bullet)) = G$ (corollary of universal covering space). Injectivity follows from the last theorem too (iff). Now if we ignore the special point, then we define the bijection as follows

$$[(\tilde{X}, p)] \rightleftarrows [p_*(\pi_1(\tilde{X}, \tilde{x_0}))]$$

for any $\tilde{x_0}$ over x_0. From the assignment, we know that the choice $\tilde{x_0}$ doesn't change the conjugacy equivalence class, moreover this map is onto by the same reason as above. Finally, we show injectivity, suppose that

$$[(p_1)_* \left(\pi_1(\tilde{X_1}, \tilde{x_1})\right)] = [(p_2)_* \left(\pi_1(\tilde{X_2}, \tilde{x_2})\right)]$$

i.e.

$$[(p_1)_*(\pi_1(\tilde{X_1}, \tilde{x_1}))] = gg^{-1}(p_2)_*(\pi_1(\tilde{X_2}, \tilde{x_2}))gg^{-1}$$

□

Example 3.16. The covering spaces of S^1 are precisely $\mathbb{R}\colon x \to e^{2\pi i x}$ and $S^1\colon z \to z^n$; these give us $\{e\}$ and $n\mathbb{Z}$ respectively.

Remark 3.17. Culture: read Deck transformations and group actions. Also read about general homotopy groups, see the pictures of the talk in class.

4 Homology.

Definition 4.1. The standard n-simplex is the set of all points in $\mathbb{R}^{n+1}$ of the form

$$\Delta_n = \left\{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} \colon t_i \geq 0 \text{ for all } i, \;\; t_0 + \cdots + t_n = 1\right\}.$$

Example 4.2.

$$\Delta_0 = \{(1)\}, \qquad \Delta_1 = \{(t_0, t_1) \colon t_0 + t_1 = 1, \; t_0, t_1 \geq 0\},$$

$$\Delta_2 = \{(t_0, t_1, t_2) \colon t_0 + t_1 + t_2 = 1, \; t_0, t_1, t_2 \geq 0\}.$$

Geometrically, Δ_0 is a point, Δ_1 is a line segment, and Δ_2 is a triangle.

Remark 4.3. Note that Δ_n can be considered as the convex hull of the standard basis vectors $e_0, \dots, e_n$ in $\mathbb{R}^{n+1}$:

$$\Delta_n = \left\{t_0 e_0 + \cdots + t_n e_n \colon t_i \geq 0, \; \sum t_i = 1\right\}.$$

More generally, if $v_0, \dots, v_n \in \mathbb{R}^{n+1}$ are affinely independent, then the convex hull of these points is homeomorphic to Δ_n by the obvious linear map sending $e_i \mapsto v_i$.

Definition 4.4. Let X be a topological space. A singular n-simplex in X is a continuous map

$$\sigma \colon \Delta_n \to X.$$

Definition 4.5. The group of singular n-chains in X, denoted $C_n(X)$, is the free abelian group generated by the set of all singular n-simplices in X.

Thus an element of $C_n(X)$ is a finite formal sum

$$n_1\sigma_1 + \cdots + n_k\sigma_k, \qquad n_i \in \mathbb{Z}.$$

Definition 4.6. Define the boundary map

$$\partial \colon C_n(X) \to C_{n-1}(X)$$

by setting $\partial = 0$ on $C_0(X)$, and for a singular n-simplex σ,

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \sigma[e_0, \dots, \hat{e_i}, \dots, e_n].$$

Here $\sigma[e_0, \dots, \hat{e_i}, \dots, e_n]$ means the restriction of σ to the ith face of Δ_n, obtained by setting $t_i = 0$.

Indeed, if $t_i = 0$, then the remaining coordinates satisfy

$$t_0 + \cdots + t_{i-1} + t_{i+1} + \cdots + t_n = 1,$$

so they define a copy of Δ_{n-1}.

Example 4.7. For 0-simplices, this is trivial since $\partial = 0$.

For a 1-simplex σ, we have

$$\partial\sigma = \sigma[\hat{e_0}, e_1] - \sigma[e_0, \hat{e_1}].$$

In other words, this is the endpoint at e_1 minus the endpoint at e_0.

For a 2-simplex,

$$\begin{aligned}\partial\sigma &= \sigma[\hat{e_0}, e_1, e_2] - \sigma[e_0, \hat{e_1}, e_2] + \sigma[e_0, e_1, \hat{e_2}] \\ &= (\text{first edge}) - (\text{second edge}) + (\text{third edge}).\end{aligned}$$

If we orient edges from the lower index to the higher index, then the signs $(-1)^i$ are exactly what make the orientations compatible.

Theorem 4.8 (Boundary of a boundary is empty)**.**

$$\partial^2 = 0.$$

Equivalently, in the sequence

$$\cdots \xrightarrow{\partial} C_{n+1}(X) \xrightarrow{\partial} C_n(X) \xrightarrow{\partial} C_{n-1}(X) \xrightarrow{\partial} \cdots$$

composing two consecutive boundary maps gives the zero homomorphism.

Proof. Let $\sigma\colon [e_0, \dots, e_n] \to X$. Then

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \sigma[e_0, \dots, \hat{e_i}, \dots, e_n].$$

Applying ∂ again,

$$\begin{aligned}
\partial(\partial\sigma) &= \sum_{i=0}^{n} (-1)^i \partial(\sigma[e_0, \dots, \hat{e_i}, \dots, e_n]) \\
&= \sum_{i=0}^{n} (-1)^i \left(\sum_{j<i} (-1)^j \sigma[e_0, \dots, \hat{e_j}, \dots, \hat{e_i}, \dots, e_n] + \sum_{j>i} (-1)^{j-1} \sigma[e_0, \dots, \hat{e_i}, \dots, \hat{e_j}, \dots, e_n] \right).
\end{aligned}$$

Every codimension-2 face appears exactly twice, once with one sign and once with the opposite sign, so all terms cancel. Hence

$$\partial^2\sigma = 0.$$

Since the singular simplices generate $C_n(X)$, it follows that $\partial^2 = 0$ on all chains. □

Remark 4.9. The above theorem says exactly that

$$\partial C_{n+1}(X) \subseteq \ker(\partial\colon C_n(X) \to C_{n-1}(X)).$$

Definition 4.10. The elements of $C_n(X)$ lying in $\ker\partial$ are called *n-cycles*.

The elements of the image of $\partial\colon C_{n+1}(X) \to C_n(X)$ are called *n-boundaries*.

Remark 4.11. Every boundary is a cycle.

Definition 4.12. The nth singular homology group of X is

$$H_n(X) = \ker\partial / \operatorname{Im}\partial.$$

That is, n-cycles modulo n-boundaries.

Example 4.13. Suppose $X = \{\cdot\}$ is a one-point space. Then for every n, there is exactly one singular n-simplex

$$\sigma\colon \Delta_n \to X.$$

Hence

$$C_n(X) \cong \mathbb{Z}$$

for all n.

Now

$$\begin{aligned}
\partial\sigma &= \sum_{i=0}^{n}(-1)^i\sigma[e_0,\ldots,\hat{e}_i,\ldots,e_n] \\
&= \left(\sum_{i=0}^{n}(-1)^i\right)\tilde{\sigma} \\
&= \begin{cases} \tilde{\sigma} & n \text{ even}, \\ 0 & n \text{ odd}, \end{cases}
\end{aligned}$$

where $\tilde{\sigma}$ denotes the unique $(n-1)$-simplex.

So the chain complex looks like

$$\cdots \xrightarrow{1} C_3(X) \xrightarrow{0} C_2(X) \xrightarrow{1} C_1(X) \xrightarrow{0} C_0(X) \to 0.$$

Therefore

$$\begin{aligned}
H_0(X) &= \mathbb{Z}/0 = \mathbb{Z}, \\
H_1(X) &= 0, \\
H_2(X) &= 0,
\end{aligned}$$

and similarly for all higher n. Hence

$$H_n(X) = \begin{cases} \mathbb{Z} & n = 0, \\ 0 & n \geq 1. \end{cases}$$

Proposition 4.14. *If X is path connected, then*

$$H_0(X) \cong \mathbb{Z}.$$

Proof. Since $\partial = 0$ on $C_0(X)$, we have

$$H_0(X) = C_0(X)/\operatorname{Im}\partial_1.$$

Define the augmentation map

$$\varepsilon\colon C_0(X) \to \mathbb{Z}$$

by

$$\varepsilon(n_1\sigma_1 + \cdots + n_k\sigma_k) = n_1 + \cdots + n_k.$$

This is surjective. We claim that

$$\ker\varepsilon = \operatorname{Im}\partial_1.$$

First, if $\sigma \in \operatorname{Im}\partial_1$, say $\sigma = \partial\tilde{\sigma}$, then σ is a sum of terms of the form endpoint minus endpoint, so the total sum of coefficients is 0. Hence

$$\operatorname{Im}\partial_1 \subseteq \ker\varepsilon.$$

Conversely, let

$$\sigma = n_1 x_1 + \cdots + n_k x_k \in C_0(X)$$

with $n_1 + \cdots + n_k = 0$. Choose a point $x_0 \in X$. Since X is path connected, for each x_i there exists a path from x_0 to x_i. Since $I \cong \Delta_1$, each such path may be viewed as a singular 1-simplex $\tilde{\sigma_i}$ with

$$\partial \tilde{\sigma_i} = x_i - x_0.$$

Therefore

$$\begin{aligned} \partial(n_1 \tilde{\sigma_1} + \cdots + n_k \tilde{\sigma_k}) &= n_1(x_1 - x_0) + \cdots + n_k(x_k - x_0) \\ &= n_1 x_1 + \cdots + n_k x_k - (n_1 + \cdots + n_k) x_0 \\ &= \sigma. \end{aligned}$$

So $\sigma \in \operatorname{Im} \partial_1$, proving

$$\ker \varepsilon = \operatorname{Im} \partial_1.$$

Hence

$$H_0(X) = C_0(X)/\operatorname{Im} \partial_1 \cong C_0(X)/\ker \varepsilon \cong \operatorname{Im} \varepsilon = \mathbb{Z}.$$

□

Proposition 4.15. *Let X have path components $\{X_\alpha\}_{\alpha \in A}$. Then*

$$H_n(X) \cong \bigoplus_{\alpha \in A} H_n(X_\alpha).$$

In particular,

$$H_0(X) \cong \bigoplus_{\alpha \in A} \mathbb{Z}.$$

Proof. Any singular n-simplex $\sigma \colon \Delta_n \to X$ must land in a single path component of X, since Δ_n is path connected. Therefore the set of singular n-simplices splits according to the path components, and so

$$C_n(X) = \bigoplus_{\alpha \in A} C_n(X_\alpha).$$

The boundary map preserves each summand, so the whole chain complex splits as a direct sum of chain complexes. Taking homology therefore gives

$$H_n(X) \cong \bigoplus_{\alpha \in A} H_n(X_\alpha).$$

The statement about H_0 follows from the previous proposition, since each path component is path connected. □

Corollary 4.16. *If X is discrete, then*

$$H_n(X) = \begin{cases} 0 & n > 0, \\ \bigoplus_{x \in X} \mathbb{Z} & n = 0. \end{cases}$$

Now suppose we have two spaces X, Y and a continuous map $f\colon X \to Y$. If $\sigma\colon \Delta_n \to X$ is an n-simplex, then

$$f \circ \sigma\colon \Delta_n \to Y$$

is also an n-simplex in Y.

Definition 4.17. Given a continuous map $f\colon X \to Y$, define

$$f_\#(\sigma) = f \circ \sigma$$

on singular simplices, and extend linearly to a homomorphism

$$f_\#\colon C_n(X) \to C_n(Y).$$

One checks that

$$f_\# \partial = \partial f_\#,$$

so we have a commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r] & C_{n+1}(X) \arrow[r, "\partial"] \arrow[d, "f_\#"'] & C_n(X) \arrow[r, "\partial"] \arrow[d, "f_\#"'] & C_{n-1}(X) \arrow[r] \arrow[d, "f_\#"'] & \cdots \\
\cdots \arrow[r] & C_{n+1}(Y) \arrow[r, "\partial"] & C_n(Y) \arrow[r, "\partial"] & C_{n-1}(Y) \arrow[r] & \cdots
\end{tikzcd}$$

Such a collection of maps commuting with the boundary operators is called a *chain map*.

More generally, a chain complex is a sequence

$$\cdots \to A_{n+1} \xrightarrow{\partial} A_n \xrightarrow{\partial} A_{n-1} \to \cdots$$

with $\partial^2 = 0$.

Remark 4.18. The identity $f_\# \partial = \partial f_\#$ follows because for $\sigma\colon \Delta_n \to X$,

$$\partial \sigma = \sum_{i=0}^{n} (-1)^i \sigma[\cdots, \hat{e}_i, \cdots],$$

so

$$f_\#(\partial \sigma) = \sum_{i=0}^{n} (-1)^i f \circ \sigma[\cdots, \hat{e}_i, \cdots],$$

while

$$\partial(f_\# \sigma) = \partial(f \circ \sigma) = \sum_{i=0}^{n} (-1)^i (f \circ \sigma)[\cdots, \hat{e}_i, \cdots].$$

Remark 4.19. Chain maps induce maps on homology:

$$f_*\colon H_n(X) \to H_n(Y), \qquad f_*([\alpha]) = [f_\#(\alpha)].$$

This is well defined. Indeed, if α is a cycle, then

$$\partial f_\#(\alpha) = f_\#(\partial\alpha) = f_\#(0) = 0,$$

so $f_\#(\alpha)$ is again a cycle. Also, if $\alpha' = \alpha + \partial\beta$, then

$$\begin{aligned} f_*([\alpha']) &= [f_\#(\alpha + \partial\beta)] \\ &= [f_\#(\alpha) + f_\#(\partial\beta)] \\ &= [f_\#(\alpha) + \partial f_\#(\beta)] \\ &= [f_\#(\alpha)]. \end{aligned}$$

Remark 4.20.
- $(1_X)_\# = 1_{C_n(X)}$.
- $(g \circ f)_\# = g_\# \circ f_\#$.

Therefore
- $(1_X)_* = 1_{H_n(X)}$.
- $(g \circ f)_* = g_* \circ f_*$.

Definition 4.21. The reduced homology groups of a space X are defined to be the homology groups of the augmented chain complex

$$\cdots \xrightarrow{\partial} C_2(X) \xrightarrow{\partial} C_1(X) \xrightarrow{\partial} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \to 0.$$

They are denoted

$$\widetilde{H}_n(X).$$

Note that

$$\widetilde{H}_n(X) = H_n(X), \qquad n > 0.$$

Remark 4.22. If X is path connected, then

$$\widetilde{H}_0(X) = 0.$$

Also

$$\widetilde{H}_n(\cdot) = 0, \qquad \forall n.$$

Theorem 4.23. *Let X, Y be topological spaces and let $f, g\colon X \to Y$ be homotopic. Then*

$$f_* = g_*\colon H_n(X) \to H_n(Y).$$

Proof. Let $H\colon X \times I \to Y$ be a homotopy from f to g.

For each singular simplex $\sigma\colon \Delta_n \to X$, consider

$$\sigma \times I\colon \Delta_n \times I \to X \times I, \qquad (\sigma \times I)(t, s) = (\sigma(t), s).$$

One subdivides $\Delta_n \times I$ into $(n+1)$ simplices in the standard prism decomposition, and defines the prism operator

$$P\colon C_n(X) \to C_{n+1}(Y)$$

by

$$P(\sigma) = \sum_{i=0}^{n} (-1)^i \, H \circ (\sigma \times I)\big[v_0, \dots, v_i, w_i, \dots, w_n\big].$$

A direct computation gives

$$\partial P + P\partial = g_\# - f_\#.$$

Now if c is a cycle, then $\partial c = 0$, so

$$g_\#(c) - f_\#(c) = (\partial P + P\partial)(c) = \partial(Pc).$$

Thus $g_\#(c)$ and $f_\#(c)$ differ by a boundary, hence define the same homology class:

$$g_*[c] = f_*[c].$$

Therefore $f_* = g_*$. □

Remark 4.24. More generally, if two chain maps are chain homotopic, then they induce the same maps on homology.

Corollary 4.25. *If $X \simeq Y$, then*

$$H_n(X) \cong H_n(Y).$$

Proof. If $f\colon X \to Y$ and $g\colon Y \to X$ are homotopy inverses, then

$$g \circ f \simeq 1_X, \qquad f \circ g \simeq 1_Y.$$

Hence

$$g_* \circ f_* = (g \circ f)_* = (1_X)_*, \qquad f_* \circ g_* = (f \circ g)_* = (1_Y)_*.$$

So f_* and g_* are inverse isomorphisms. □

Corollary 4.26. *If X is contractible, then*

$$H_n(X) = \begin{cases} \mathbb{Z} & n = 0, \\ 0 & n > 0. \end{cases}$$

4.1 Second Algebraic Detour

Definition 4.27. An exact sequence of abelian groups is a sequence

$$\cdots \to A_{n+1} \xrightarrow{f_{n+1}} A_n \xrightarrow{f_n} A_{n-1} \to \cdots$$

such that

$$\operatorname{Im} f_{n+1} = \ker f_n$$

for all n.

A short exact sequence is one of the form

$$0 \to A \to B \to C \to 0.$$

Example 4.28. Many algebraic statements can be expressed in terms of exactness:

- $0 \to A \xrightarrow{f} B$ is exact iff $\ker f = 0$, i.e. iff f is injective.
- $A \xrightarrow{f} B \to 0$ is exact iff f is surjective.
- $0 \to A \xrightarrow{f} B \to 0$ is exact iff f is an isomorphism.
- $0 \to A \to B \to C \to 0$ is exact iff $A \to B$ is injective, $B \to C$ is surjective, and

$$B/A \cong C.$$

- If $A \subset B \subset C$, then

$$0 \to B/A \to C/A \to C/B \to 0$$

  is exact.

Definition 4.29. A short exact sequence of chain complexes is a diagram

$$0 \to A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{q} C_\bullet \to 0$$

such that for each n the sequence

$$0 \to A_n \xrightarrow{i_n} B_n \xrightarrow{q_n} C_n \to 0$$

is exact, and the maps i, q commute with the boundary maps.

Theorem 4.30. *Suppose we have a short exact sequence of chain complexes*

$$0 \to A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{q} C_\bullet \to 0.$$

Then there is a long exact sequence in homology

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{q_*} H_n(C) \xrightarrow{\delta} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \to \cdots$$

where δ is the connecting homomorphism.

Proof. We first define δ. Let $[c] \in H_n(C)$, so $c \in C_n$ is a cycle. Since $q\colon B_n \to C_n$ is surjective, choose $b \in B_n$ with $q(b) = c$. Then

$$q(\partial b) = \partial q(b) = \partial c = 0,$$

so $\partial b \in \ker q = \operatorname{Im} i$. Hence there exists $a \in A_{n-1}$ with

$$i(a) = \partial b.$$

One checks that a is a cycle, and defines

$$\delta([c]) = [a].$$

This is well defined.

The exactness of the resulting long sequence is proved by diagram chasing. For example: At $H_n(B)$, we show

$$\ker q_* = \operatorname{Im} i_*.$$

Since $q \circ i = 0$, certainly $\operatorname{Im} i_* \subseteq \ker q_*$. Conversely, if $[b] \in \ker q_*$, then $[q(b)] = 0$, so $q(b) = \partial c$ for some $c \in C_{n+1}$. Lift c to some $\beta \in B_{n+1}$ with $q(\beta) = c$. Then

$$q(b - \partial\beta) = q(b) - \partial q(\beta) = q(b) - \partial c = 0,$$

so $b - \partial\beta \in \ker q = \operatorname{Im} i$. Thus there exists $a \in A_n$ with

$$i(a) = b - \partial\beta.$$

Since b is a cycle, one checks that a is a cycle and

$$i_*([a]) = [b].$$

Hence $\ker q_* \subseteq \operatorname{Im} i_*$.

The exactness at the other terms is similar. $\square$

4.2 Relative Homology

Definition 4.31. Let $A \subset X$. Since every singular simplex in A is also a singular simplex in X, we have

$$C_n(A) \subseteq C_n(X).$$

Moreover, the boundary map sends $C_n(A)$ into $C_{n-1}(A)$, so it descends to the quotient

$$C_n(X, A) := C_n(X)/C_n(A).$$

These are called the *relative n-chains*. The induced boundary map again satisfies $\partial^2 = 0$, so we obtain the chain complex

$$\cdots \to C_{n+1}(X, A) \xrightarrow{\partial} C_n(X, A) \xrightarrow{\partial} C_{n-1}(X, A) \to \cdots.$$

Its homology groups are called the *relative homology groups* of the pair (X, A), and are denoted

$$H_n(X, A).$$

Now we have a short exact sequence of chain complexes

$$0 \to C_n(A) \to C_n(X) \to C_n(X, A) \to 0.$$

Therefore we get a long exact sequence in homology:

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X,A) \xrightarrow{\delta} H_{n-1}(A) \to \cdots.$$

We can also extend the augmented chain complexes and obtain the long exact sequence in reduced homology. In particular,

$$\widetilde{H}_n(X,A) = H_n(X,A).$$

Corollary 4.32. *If* $A = \{\cdot\}$, *then*

$$H_n(X,\cdot) \cong \widetilde{H}_n(X).$$

Proof. Since $\widetilde{H}_n(\cdot) = 0$ for all n, the long exact sequence gives isomorphisms

$$\widetilde{H}_n(X) \cong H_n(X,\cdot).$$

□

Now suppose $A \subset U \subset X$. Then there is a short exact sequence of chain complexes

$$0 \to C_n(U,A) \to C_n(X,A) \to C_n(X,U) \to 0.$$

Hence we get a long exact sequence

$$\cdots \to H_n(U,A) \to H_n(X,A) \to H_n(X,U) \to H_{n-1}(U,A) \to \cdots.$$

Definition 4.33. A map of pairs

$$(X,A) \to (Y,B)$$

is a continuous map $f\colon X \to Y$ such that $f(A) \subseteq B$.

Remark 4.34. In this situation, $f_\#$ sends $C_n(X)$ to $C_n(Y)$ and $C_n(A)$ to $C_n(B)$, so it descends to a well-defined map

$$f_\#\colon C_n(X,A) \to C_n(Y,B)$$

given by

$$f_\#[c] = [f_\# c].$$

Since $\partial f_\# = f_\# \partial$, this induces a map

$$f_*\colon H_n(X,A) \to H_n(Y,B).$$

Now suppose that $f, g\colon (X,A) \to (Y,B)$ are maps of pairs and that $f \simeq g$ by a homotopy

$$H\colon X \times I \to Y$$

such that for every $t \in I$, the map H_t sends A into B. Then the prism operator still makes sense and sends $C_n(A)$ into $C_{n+1}(B)$. Hence it descends to the quotient and gives

$$\partial P + P\partial = g_\# - f_\#$$

on relative chains. Therefore

$$g_* = f_* \colon H_n(X, A) \to H_n(Y, B).$$

Theorem 4.35 (Excision)**.** *Suppose $A \subset U \subset X$ with A closed and U open. Then the inclusion*

$$i \colon (X \setminus A,\, U \setminus A) \hookrightarrow (X, U)$$

induces an isomorphism

$$H_n(X \setminus A,\, U \setminus A) \cong H_n(X, U).$$

Equivalently, if $X = U \cup V$ with U, V open, then the inclusion

$$i \colon (V, U \cap V) \hookrightarrow (X, U)$$

induces an isomorphism

$$H_n(V, U \cap V) \cong H_n(X, U).$$

Definition 4.36. $C_n(U + V)$ is the free abelian group generated by singular simplices whose image lies either in U or in V.

Lemma 4.37. *The inclusion*

$$i \colon C_n(U + V) \hookrightarrow C_n(X)$$

induces an isomorphism

$$H_n(U + V) \cong H_n(X).$$

Similarly, the induced map

$$C_n(U + V)/C_n(U) \hookrightarrow C_n(X)/C_n(U)$$

induces an isomorphism

$$H_n(U + V, U) \cong H_n(X, U).$$

Proof. This is proved using barycentric subdivision. See Hatcher. □

Proof of Excision. First note that the two formulations are equivalent by taking $V = X \setminus A$, so that

$$U \cap V = U \setminus A.$$

By the lemma, the inclusion induces an isomorphism

$$H_n(U + V, U) \cong H_n(X, U).$$

Now consider the inclusion

$$C_n(V) \hookrightarrow C_n(U+V).$$

It sends $C_n(U \cap V)$ into $C_n(U)$, so it descends to a map

$$C_n(V)/C_n(U \cap V) \to C_n(U+V)/C_n(U).$$

This map is actually an isomorphism, because singular simplices in V split into those landing in $U \cap V$ and those not contained in U, and the same description applies to simplices in $U+V$ modulo those in U. Hence

$$H_n(V, U \cap V) \cong H_n(U+V, U) \cong H_n(X, U).$$

□

Definition 4.38. Suppose we have $A \subset U \subset X$ and A closed with U open. If A is a deformation retract of U, then we say that (X, A) is a good pair.

Theorem 4.39. *Suppose that (X, A) is a good pair, then the quotient map*

$$q\colon (X, A) \to (X/A, A/A)$$

Then this quotient map induces an isomorphism of homology groups

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$$

Proof. Since (X, A) is a good pair, there exists an open set U with $A \subset U \subset X$ such that A is a deformation retract of U.

Consider the quotient map

$$q : (X, A) \to (X/A, A/A).$$

Write $*$ for the point $A/A \in X/A$, and write U/A for the image of U in X/A.

Since A deformation retracts onto U, we have

$$H_n(U, A) = 0$$

for all n. Therefore, from the long exact sequence of the triple

$$A \subset U \subset X,$$

the map

$$H_n(X, A) \to H_n(X, U)$$

is an isomorphism.

Similarly, since U/A deformation retracts onto the point $A/A = *$, we have

$$H_n(U/A, *) = 0$$

for all n. Hence the long exact sequence of the triple

$$* \subset U/A \subset X/A$$

gives an isomorphism

$$H_n(X/A, *) \to H_n(X/A, U/A).$$

Now compare $H_n(X, U)$ and $H_n(X/A, U/A)$. Since $A \subset U$ and U is open, excision allows us to remove A from the pair (X, U):

$$H_n(X, U) \cong H_n(X - A, U - A).$$

Also, the quotient map restricts to a homeomorphism

$$X - A \cong (X/A) - *,$$

and under this homeomorphism,

$$U - A \cong (U/A) - *.$$

Therefore

$$H_n(X - A, U - A) \cong H_n((X/A) - *, (U/A) - *).$$

By excision again, removing the point $*$ from the pair $(X/A, U/A)$ gives

$$H_n((X/A) - *, (U/A) - *) \cong H_n(X/A, U/A).$$

Thus we have a commutative diagram

$$\begin{array}{ccc} H_n(X, A) & \longrightarrow & H_n(X, U) \\ \downarrow q_* & & \downarrow\cong \\ H_n(X/A, *) & \longrightarrow & H_n(X/A, U/A), \end{array}$$

where the horizontal maps are isomorphisms and the right vertical map is an isomorphism by excision. Hence the left vertical map is also an isomorphism:

$$H_n(X, A) \cong H_n(X/A, A/A).$$

Finally, since A/A is a single point, we have

$$H_n(X/A, A/A) \cong \widetilde{H}_n(X/A).$$

Therefore
$$H_n(X, A) \cong H_n(X/A, A/A) \cong \widetilde{H}_n(X/A).$$

□

Corollary 4.40. *If (X, A) is a good pair, we have the long exact sequence of homology groups*

$$H_n(A) \to H_n(X) \to H_n(X, A) \to H_{n-1}(A) \to \cdots$$

Which gives us
$$\tilde{H}_n(A) \to \tilde{H}_n(X) \to \tilde{H}_n(X/A) \to \tilde{H}_{n-1}(A) \to \cdots$$

Corollary 4.41. $\tilde{H}_n(S^k) = \begin{cases} 0 & n \neq k \\ \mathbb{Z} & n = k \end{cases}$

Corollary 4.42. *For every $k \geq 0$,*

$$\widetilde{H}_n(S^k) \cong \begin{cases} \mathbb{Z} & n = k, \\ 0 & n \neq k. \end{cases}$$

Proof. Let $X = D^k$ and $A = \partial D^k = S^{k-1}$. This is a good pair, and

$$X/A = D^k/\partial D^k \cong S^k.$$

Therefore the long exact sequence of the pair gives

$$\cdots \to \widetilde{H}_n(S^{k-1}) \to \widetilde{H}_n(D^k) \to \widetilde{H}_n(S^k) \to \widetilde{H}_{n-1}(S^{k-1}) \to \widetilde{H}_{n-1}(D^k) \to \cdots$$

Since D^k is contractible, we have

$$\widetilde{H}_i(D^k) = 0 \qquad \text{for all } i.$$

Hence the connecting homomorphism gives an isomorphism

$$\widetilde{H}_n(S^k) \cong \widetilde{H}_{n-1}(S^{k-1}).$$

Iterating this isomorphism, we get

$$\widetilde{H}_n(S^k) \cong \widetilde{H}_{n-k}(S^0).$$

Now S^0 consists of two points. We compute its reduced homology.
Since $S^0 = \{p, q\}$ is discrete, there are no nonconstant singular 1-simplices. Thus

$$\operatorname{Im}(\partial_1) = 0.$$

Also
$$C_0(S^0) \cong \mathbb{Z}p \oplus \mathbb{Z}q \cong \mathbb{Z}^2.$$

The augmentation map is

$$\varepsilon : C_0(S^0) \to \mathbb{Z}, \qquad \varepsilon(ap + bq) = a + b.$$

Therefore

$$\widetilde{H}_0(S^0) = \ker \varepsilon = \{ap + bq : a + b = 0\} = \langle p - q \rangle \cong \mathbb{Z}.$$

Also $\widetilde{H}_i(S^0) = 0$ for $i \neq 0$.

Therefore

$$\widetilde{H}_n(S^k) \cong \widetilde{H}_{n-k}(S^0) \cong \begin{cases} \mathbb{Z} & n - k = 0, \\ 0 & n - k \neq 0. \end{cases}$$

Hence

$$\widetilde{H}_n(S^k) \cong \begin{cases} \mathbb{Z} & n = k, \\ 0 & n \neq k. \end{cases}$$

□

Proposition 4.43. *If X is nonempty, then*

$$H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}.$$

Moreover, both $H_0(X)$ and $\widetilde{H}_0(X)$ are free abelian groups.

Proof. Let $\pi_0(X)$ denote the set of path components of X. For each path component $P \in \pi_0(X)$, choose a point

$$x_P \in P.$$

We first recall the standard computation of $H_0(X)$. The group $H_0(X)$ is the free abelian group on the path components of X. More precisely,

$$H_0(X) \cong \bigoplus_{P \in \pi_0(X)} \mathbb{Z}[x_P].$$

Indeed, if two points $x, y \in X$ lie in the same path component, then there is a path $\gamma : I \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$. Viewing γ as a singular 1-simplex, we have

$$\partial \gamma = y - x.$$

Thus

$$[x] = [y] \in H_0(X).$$

Therefore every point in a path component represents the same class in $H_0(X)$.

On the other hand, if two points lie in different path components, there is no singular 1-chain whose boundary identifies them. Hence the distinct path components give independent generators. Therefore

$$H_0(X) \cong \bigoplus_{P \in \pi_0(X)} \mathbb{Z}[x_P].$$

Now consider the augmentation map

$$\bar{\varepsilon} : H_0(X) \to \mathbb{Z},$$

defined by

$$\bar{\varepsilon}\left(\sum_{P\in\pi_0(X)} n_P[x_P]\right) = \sum_{P\in\pi_0(X)} n_P.$$

Then

$$\widetilde{H}_0(X) = \ker \bar{\varepsilon}.$$

Thus

$$\widetilde{H}_0(X) = \left\{ \sum_{P\in\pi_0(X)} n_P[x_P] \;\middle|\; \sum_{P\in\pi_0(X)} n_P = 0 \right\}.$$

Since X is nonempty, choose one distinguished path component $P_0 \in \pi_0(X)$. We claim that

$$\widetilde{H}_0(X) \cong \bigoplus_{P\neq P_0} \mathbb{Z}\big([x_P] - [x_{P_0}]\big).$$

Indeed, let

$$z = \sum_{P\in\pi_0(X)} n_P[x_P] \in \widetilde{H}_0(X).$$

Then

$$\sum_{P\in\pi_0(X)} n_P = 0.$$

Hence

$$n_{P_0} = -\sum_{P\neq P_0} n_P.$$

Therefore

$$z = \sum_{P\neq P_0} n_P[x_P] + n_{P_0}[x_{P_0}] = \sum_{P\neq P_0} n_P[x_P] - \sum_{P\neq P_0} n_P[x_{P_0}],$$

so

$$z = \sum_{P\neq P_0} n_P\big([x_P] - [x_{P_0}]\big).$$

This expression is unique, because the classes $[x_P]$ form a basis for $H_0(X)$. Therefore $\widetilde{H}_0(X)$ is free abelian with basis

$$\{[x_P] - [x_{P_0}] \mid P \in \pi_0(X),\ P \neq P_0\}.$$

Now define

$$\Phi : \widetilde{H}_0(X) \oplus \mathbb{Z} \to H_0(X)$$

by

$$\Phi(z, m) = z + m[x_{P_0}].$$

We show that Φ is an isomorphism.

First, Φ is surjective. Let

$$w = \sum_{P \in \pi_0(X)} n_P [x_P] \in H_0(X).$$

Set

$$m = \sum_{P \in \pi_0(X)} n_P.$$

Then

$$w - m[x_{P_0}] \in \ker \overline{\varepsilon} = \widetilde{H}_0(X),$$

because

$$\overline{\varepsilon}(w - m[x_{P_0}]) = \overline{\varepsilon}(w) - m = m - m = 0.$$

Thus

$$w = (w - m[x_{P_0}]) + m[x_{P_0}] = \Phi(w - m[x_{P_0}], m).$$

Hence Φ is surjective.

Second, Φ is injective. Suppose

$$\Phi(z, m) = 0.$$

Then

$$z + m[x_{P_0}] = 0.$$

Applying $\overline{\varepsilon}$ gives

$$\overline{\varepsilon}(z) + m\overline{\varepsilon}([x_{P_0}]) = 0.$$

Since $z \in \widetilde{H}_0(X) = \ker \overline{\varepsilon}$ and

$$\overline{\varepsilon}([x_{P_0}]) = 1,$$

we get

$$0 + m = 0.$$

Hence $m = 0$. Therefore $z = 0$. So Φ is injective.

Thus

$$H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}.$$

Moreover, from the explicit bases above, both $H_0(X)$ and $\widetilde{H}_0(X)$ are free abelian groups. □

4.3 Mayer-Vietoris Sequences

Suppose that $X = U \cup V$, where U, V are open in X.

Define $C_n(U + V)$ to be the subgroup of $C_n(X)$ generated by singular simplices whose image lies entirely in either U or V. Thus

$$C_n(U + V) = C_n(U) + C_n(V) \subseteq C_n(X).$$

We have inclusions

$$C_n(U) \hookrightarrow C_n(U + V), \qquad C_n(V) \hookrightarrow C_n(U + V).$$

Therefore we get a chain map

$$C_n(U) \oplus C_n(V) \to C_n(U+V), \qquad (\sigma, \tau) \mapsto \sigma + \tau.$$

This map is surjective by definition of $C_n(U+V)$.

We also have inclusions

$$C_n(U \cap V) \hookrightarrow C_n(U), \qquad C_n(U \cap V) \hookrightarrow C_n(V).$$

Define

$$C_n(U \cap V) \to C_n(U) \oplus C_n(V), \qquad \sigma \mapsto (\sigma, -\sigma).$$

This is an injective chain map, and its image is the kernel of the previous map. Therefore we have a short exact sequence of chain complexes

$$0 \to C_*(U \cap V) \to C_*(U) \oplus C_*(V) \to C_*(U+V) \to 0.$$

Hence we get a long exact sequence in homology:

$$\cdots \to H_n(U \cap V) \to H_n(U) \oplus H_n(V) \to H_n(U+V) \to H_{n-1}(U \cap V) \to \cdots .$$

By barycentric subdivision, the inclusion

$$C_*(U+V) \hookrightarrow C_*(X)$$

induces an isomorphism on homology. Therefore

$$H_n(U+V) \cong H_n(X).$$

Thus the Mayer-Vietoris sequence becomes

$$\cdots \to H_n(U \cap V) \to H_n(U) \oplus H_n(V) \to H_n(X) \to H_{n-1}(U \cap V) \to \cdots .$$

Similarly, using the augmented chain complexes, we obtain the reduced Mayer-Vietoris sequence:

$$\cdots \to \widetilde{H}_n(U \cap V) \to \widetilde{H}_n(U) \oplus \widetilde{H}_n(V) \to \widetilde{H}_n(X) \to \widetilde{H}_{n-1}(U \cap V) \to \cdots .$$

Theorem 4.44 (Brouwer Fixed Point Theorem)**.** *Every continuous map*

$$f : D^n \to D^n$$

has a fixed point. That is, there exists $x \in D^n$ such that

$$f(x) = x.$$

Proof. Suppose, for contradiction, that f has no fixed point. Thus

$$f(x) \neq x \qquad \text{for every } x \in D^n.$$

For each $x \in D^n$, consider the ray starting at $f(x)$ and passing through x. Since $f(x) \neq x$, this ray is well-defined. Let $r(x)$ be the point where this ray intersects the boundary sphere $S^{n-1} = \partial D^n$.

This defines a continuous map

$$r : D^n \to S^{n-1}.$$

Moreover, if $x \in S^{n-1}$, then the ray starting at $f(x)$ and passing through x intersects the boundary at x itself. Hence

$$r(x) = x \qquad \text{for every } x \in S^{n-1}.$$

Therefore r is a retraction of D^n onto S^{n-1}.

Let

$$i : S^{n-1} \hookrightarrow D^n$$

be the inclusion. Since r is a retraction, we have

$$r \circ i = \mathrm{id}_{S^{n-1}}.$$

Applying reduced homology in degree $n-1$, we get

$$r_* \circ i_* = (\mathrm{id}_{S^{n-1}})_* = \mathrm{id}_{\widetilde{H}_{n-1}(S^{n-1})}.$$

But

$$\widetilde{H}_{n-1}(D^n) = 0$$

because D^n is contractible, while

$$\widetilde{H}_{n-1}(S^{n-1}) \cong \mathbb{Z}.$$

Hence i_* maps into the zero group, so

$$r_* \circ i_* = 0.$$

This contradicts the fact that

$$r_* \circ i_* = \mathrm{id}_{\widetilde{H}_{n-1}(S^{n-1})}.$$

Therefore f must have a fixed point. □

Theorem 4.45 (Invariance of Dimension). *Suppose $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ are open subsets. If*

$$f : U \to V$$

is a homeomorphism, then

$$n = m.$$

Proof. Choose $x_0 \in U$. Since U is open, there is an open ball around x_0 contained in U.

We claim that
$$H_k(U, U - \{x_0\}) \cong \widetilde{H}_{k-1}(S^{n-1}).$$
Equivalently, this group is $\mathbb{Z}$ when $k = n$ and 0 otherwise.

Indeed, by excision,
$$H_k(U, U - \{x_0\}) \cong H_k(\mathbb{R}^n, \mathbb{R}^n - \{x_0\}).$$
Since $\mathbb{R}^n$ is contractible, the long exact sequence of the pair gives
$$H_k(\mathbb{R}^n, \mathbb{R}^n - \{x_0\}) \cong \widetilde{H}_{k-1}(\mathbb{R}^n - \{x_0\}).$$
But
$$\mathbb{R}^n - \{x_0\} \simeq S^{n-1}.$$
Therefore
$$H_k(U, U - \{x_0\}) \cong \widetilde{H}_{k-1}(S^{n-1}).$$
Hence
$$H_k(U, U - \{x_0\}) \cong \begin{cases} \mathbb{Z} & k = n, \\ 0 & k \neq n. \end{cases}$$

Now f is a homeomorphism, so it induces a homeomorphism of pairs
$$(U, U - \{x_0\}) \cong (V, V - \{f(x_0)\}).$$
Therefore
$$H_k(U, U - \{x_0\}) \cong H_k(V, V - \{f(x_0)\})$$
for every k.

Applying the computation above to $V \subseteq \mathbb{R}^m$, we get
$$H_k(V, V - \{f(x_0)\}) \cong \begin{cases} \mathbb{Z} & k = m, \\ 0 & k \neq m. \end{cases}$$
Therefore the only degree in which the local homology group is nonzero is n on the left and m on the right. Hence
$$n = m.$$
□

Theorem 4.46 (Jordan Curve Theorem)**.** *Suppose $C \subset S^2$ and*
$$C \cong S^1.$$
Then $S^2 - C$ has exactly two path components.

Proof. It is enough to prove that

$$\widetilde{H}_0(S^2 - C) \cong \mathbb{Z}.$$

Indeed, if a space has r path components, then

$$\widetilde{H}_0(X) \cong \mathbb{Z}^{r-1}.$$

Thus $\widetilde{H}_0(S^2 - C) \cong \mathbb{Z}$ implies that $S^2 - C$ has exactly two path components.

We first prove the following lemma.

Lemma. If $E \subset S^2$ is homeomorphic to the interval $I = [0, 1]$, then

$$\widetilde{H}_k(S^2 - E) = 0 \qquad \text{for all } k.$$

Proof of the lemma. Let

$$\varphi : [0, 1] \to E$$

be a homeomorphism.

Suppose, for contradiction, that there exists a nonzero class

$$[z] \in \widetilde{H}_k(S^2 - E).$$

Split the interval into two closed subintervals:

$$I_1 = \left[0, \frac{1}{2}\right], \qquad I_2 = \left[\frac{1}{2}, 1\right].$$

Let

$$A = \varphi(I_1), \qquad B = \varphi(I_2).$$

Define

$$U = S^2 - A, \qquad V = S^2 - B.$$

Then U and V are open subsets of S^2, and

$$U \cap V = S^2 - E,$$

while

$$U \cup V = S^2 - \{\varphi(1/2)\}.$$

Since

$$S^2 - \{\varphi(1/2)\} \cong \mathbb{R}^2,$$

we have

$$\widetilde{H}_j(U \cup V) = 0 \qquad \text{for all } j.$$

The reduced Mayer-Vietoris sequence gives

$$\cdots \to \widetilde{H}_{k+1}(U \cup V) \to \widetilde{H}_k(U \cap V) \to \widetilde{H}_k(U) \oplus \widetilde{H}_k(V) \to \widetilde{H}_k(U \cup V) \to \cdots.$$

Since the outer groups are zero, we get an isomorphism

$$\widetilde{H}_k(S^2 - E) \cong \widetilde{H}_k(U) \oplus \widetilde{H}_k(V).$$

Therefore the nonzero class $[z]$ remains nonzero in at least one of

$$\widetilde{H}_k(S^2 - A) \qquad \text{or} \qquad \widetilde{H}_k(S^2 - B).$$

Hence we can replace E by one of its two halves while preserving the property that $[z]$ is nonzero in the complement. Repeating this construction, we obtain a nested sequence of closed intervals

$$J_1 \supset J_2 \supset J_3 \supset \cdots$$

such that the length of J_i tends to 0, and such that $[z]$ is nonzero in

$$\widetilde{H}_k(S^2 - \varphi(J_i))$$

for every i.

Since the intervals are nested and their lengths tend to 0, their intersection is a single point:

$$\bigcap_i J_i = \{t_0\}.$$

Therefore

$$\bigcap_i \varphi(J_i) = \{\varphi(t_0)\}.$$

But

$$S^2 - \{\varphi(t_0)\} \cong \mathbb{R}^2,$$

so

$$\widetilde{H}_k(S^2 - \{\varphi(t_0)\}) = 0.$$

Hence the cycle z bounds in $S^2 - \{\varphi(t_0)\}$. Thus there exists a chain w in $S^2 - \{\varphi(t_0)\}$ such that

$$\partial w = z.$$

The chain w is a finite linear combination of singular simplices, so the union of the images of its simplices is compact. This compact set does not contain $\varphi(t_0)$. Since the sets

$$S^2 - \varphi(J_i)$$

form an increasing open cover of this compact set, the compact set is contained in $S^2 - \varphi(J_N)$ for some N.

Therefore w is actually a chain in $S^2 - \varphi(J_N)$, and

$$\partial w = z.$$

This contradicts the choice of J_N, since $[z]$ was supposed to be nonzero in

$$\widetilde{H}_k(S^2 - \varphi(J_N)).$$

Therefore

$$\widetilde{H}_k(S^2 - E) = 0 \qquad \text{for all } k.$$

This proves the lemma.

Now let

$$\varphi : S^1 \to C$$

be a homeomorphism. Write S^1 as the union of two closed arcs:

$$S^1 = A \cup B,$$

where

$$A \cap B = \{p, q\}.$$

Then $\varphi(A)$ and $\varphi(B)$ are both homeomorphic to intervals.

Define

$$U = S^2 - \varphi(A), \qquad V = S^2 - \varphi(B).$$

Then U and V are open in S^2, and

$$U \cap V = S^2 - C.$$

Also,

$$U \cup V = S^2 - \varphi(A \cap B) = S^2 - \{\varphi(p), \varphi(q)\}.$$

By the lemma,

$$\widetilde{H}_k(U) = 0 \qquad \text{and} \qquad \widetilde{H}_k(V) = 0$$

for all k.

The reduced Mayer-Vietoris sequence gives

$$\cdots \to \widetilde{H}_{k+1}(U) \oplus \widetilde{H}_{k+1}(V) \to \widetilde{H}_{k+1}(U \cup V) \to \widetilde{H}_k(U \cap V) \to \widetilde{H}_k(U) \oplus \widetilde{H}_k(V) \to \cdots .$$

Since the homology groups of U and V vanish, we get an isomorphism

$$\widetilde{H}_k(U \cap V) \cong \widetilde{H}_{k+1}(U \cup V).$$

Therefore

$$\widetilde{H}_k(S^2 - C) \cong \widetilde{H}_{k+1}(S^2 - \{\varphi(p), \varphi(q)\}).$$

But

$$S^2 - \{\varphi(p), \varphi(q)\} \simeq S^1.$$

Hence

$$\widetilde{H}_k(S^2 - C) \cong \widetilde{H}_{k+1}(S^1).$$

Taking $k = 0$, we get

$$\widetilde{H}_0(S^2 - C) \cong \widetilde{H}_1(S^1) \cong \mathbb{Z}.$$

Therefore $S^2 - C$ has exactly two path components. □

Theorem 4.47. *Let X be path connected, and let $x_0 \in X$. Then*

$$\pi_1(X, x_0)^{\mathrm{ab}} \cong H_1(X).$$

Equivalently,

$$\pi_1(X, x_0)/[\pi_1(X, x_0), \pi_1(X, x_0)] \cong H_1(X).$$

Proof. Let

$$G = \pi_1(X, x_0).$$

We define a map

$$\psi : G \longrightarrow H_1(X)$$

by sending a based loop to the homology class of the same loop regarded as a singular 1-cycle.

More explicitly, if

$$\gamma : I \to X$$

is a loop based at x_0, then as a singular 1-simplex it satisfies

$$\partial\gamma = \gamma(1) - \gamma(0) = x_0 - x_0 = 0.$$

So γ is a 1-cycle. Define

$$\psi([\gamma]) = [\gamma] \in H_1(X).$$

First, this is well-defined. If γ_0 and γ_1 are homotopic rel endpoints, then the homotopy gives a singular square in X. Divide the square into two triangles. The boundary of this 2-chain is

$$\gamma_1 - \gamma_0$$

plus two constant paths at x_0. Constant paths are boundaries, so they do not matter in homology. Hence

$$[\gamma_0] = [\gamma_1] \in H_1(X).$$

Therefore ψ is well-defined.

Next, ψ is a homomorphism. If γ_1, γ_2 are based loops, then geometrically the loop

$$\gamma_1 * \gamma_2$$

is homologous to the sum of the two loops

$$\gamma_1 + \gamma_2.$$

Indeed, look at the standard 2-simplex whose boundary consists of the edge γ_1, then the edge γ_2, and then the reverse of the concatenated edge $\gamma_1 * \gamma_2$. Its boundary is

$$\gamma_1 + \gamma_2 - (\gamma_1 * \gamma_2).$$

So

$$[\gamma_1 * \gamma_2] = [\gamma_1] + [\gamma_2] \in H_1(X).$$

Thus

$$\psi([\gamma_1][\gamma_2]) = \psi([\gamma_1]) + \psi([\gamma_2]).$$

Since $H_1(X)$ is abelian, every commutator in G maps to 0. Hence

$$[G, G] \subseteq \ker \psi.$$

Therefore ψ descends to a homomorphism

$$\overline{\psi} : G^{\mathrm{ab}} \longrightarrow H_1(X).$$

We now show that this map is an isomorphism.

First, $\overline{\psi}$ is surjective. Let z be a singular 1-cycle. Write

$$z = \sum_i n_i \sigma_i,$$

where each σ_i is a singular 1-simplex. Since z is a cycle,

$$\partial z = 0.$$

Geometrically this means that all endpoints cancel. So the edges in z can be arranged into closed paths: whenever one edge ends at a point, another edge begins there, counting multiplicity and orientation.

Thus z is homologous to a sum of loops in X.

Now these loops may not be based at x_0. But since X is path connected, for every point $x \in X$ we may choose a path

$$\eta_x$$

from x_0 to x. If γ is a loop based at x, then

$$\eta_x * \gamma * \overline{\eta_x}$$

is a loop based at x_0.

Also, γ is homologous to this based loop. Indeed, the extra path η_x and its reverse $\overline{\eta_x}$ cancel in homology:

$$[\eta_x * \gamma * \overline{\eta_x}] = [\eta_x] + [\gamma] - [\eta_x] = [\gamma].$$

Therefore every 1-cycle is homologous to a sum of based loops. Hence every class in $H_1(X)$ is in the image of $\overline{\psi}$, so $\overline{\psi}$ is surjective.

It remains to show injectivity. Suppose a based loop γ maps to zero in $H_1(X)$. This means that γ is the boundary of some singular 2-chain:

$$\gamma = \partial c.$$

So geometrically we have a collection of singular 2-simplices whose oriented boundary edges cancel in pairs, except for the outer boundary, which is γ.

Look at this collection of triangles. Each triangle has boundary a loop made of three edges:

$$\sigma_1 + \sigma_2 - \sigma_3.$$

In the fundamental group, after choosing paths from x_0 to the vertices, this triangle gives the relation

$$\sigma_1 \sigma_2 \sigma_3^{-1} = 1.$$

Equivalently,

$$\sigma_3 = \sigma_1 \sigma_2.$$

So each triangle tells us that one side of the triangle is homotopic, after passing to the abelianized fundamental group, to the product of the other two sides.

Now multiply together the relations coming from all the triangles in the 2-chain. Every interior edge appears twice, once with each orientation. Therefore these interior edges cancel. The only edges left are the boundary edges, namely the loop γ.

But in the abelianization, the order of multiplication does not matter. So after all the pairwise cancellations, the product of the triangle-boundary relations gives

$$[\gamma] = 1 \in G^{\mathrm{ab}}.$$

Thus any based loop which is null-homologous maps to the identity in the abelianization.

Hence

$$\ker \overline{\psi} = 0.$$

So $\overline{\psi}$ is injective.

Therefore $\overline{\psi}$ is both surjective and injective, and hence is an isomorphism:

$$G^{\mathrm{ab}} \cong H_1(X).$$

Since $G = \pi_1(X, x_0)$, we get

$$\pi_1(X, x_0)^{\mathrm{ab}} \cong H_1(X).$$

Equivalently,

$$\pi_1(X, x_0)/[\pi_1(X, x_0), \pi_1(X, x_0)] \cong H_1(X).$$

□